Using Data to Drive Business Performance

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Nimrod Dvir, PhD · Publisher discussion draft · October 2026

What is this book about?

This is courseware about the relationship between data and business performance. Students learn how the way they organize, query and present data changes the decisions a business can make.

I want students to see the whole journey: understand the data, build a reliable database, ask a useful question in SQL, check the answer, and turn it into a recommendation. AI is part of that journey, with students responsible for checking and explaining the work.

The course arc

Data → Tables → Relationships → Queries → Analytics → Decisions

A familiar rhythm in every chapter

The next edition is organized into **16 chapters in five parts** . Every chapter combines the following components, and a semester project grows across three checkpoints. The existing **Grading Database** supports guided practice; **PetVax** lets students try the same skills independently. We can review both cases together.

- **Introduction.** The business question and the chapter objectives.
- **Core concepts.** The main chapter, in numbered sections with worked examples.
- **Let's Build.** Guided, hands-on practice on the running case.
- **Review questions.** Recall and reflection.
- **Reading Assessment Test.** An auto-graded quiz, tied to the book's own databases.
- **Interactive lab.** Independent practice on a second case, with hints and verification tasks.

• **Terms Treasury.** The key terms, defined in context.

• **Key takeaways.** A summary and a bridge to the next chapter.

Instructor materials

The courseware also includes materials for instructors:

Review questions · Terms Treasury · Answer keys · Automatic quiz grading · Rubrics · Slides · Lesson plans · Instructor dashboard

Who it is for

Primary course	The applied business database or MIS data-management course: modeling, normalization, SQL through window functions, administration, warehousing and business intelligence, taught through business cases.
Business core	The required data or information systems course for accounting, finance, marketing, management and operations students, who need to query, check and present data. A shorter route through the same chapters serves it.
Adjacent courses	Accounting information systems, business analytics foundations, and bridge courses in IS master's programs.

Why and how it was built

After seven years of teaching this course to more than 4,000 students, I had a familiar problem: I liked parts of several textbooks, but no single book brought together everything I wanted students to learn. I often assigned two books and wrote the missing pieces myself.

So I began building the courseware I wanted to teach. Over a year of writing and work with colleagues, it became a complete course: a small student test in summer 2026, followed by version 2 in use with 500 students in fall 2026.

The classroom gave me the starting point. The research helped me question and refine it.

What informed the design

• **Teaching:** seven years of seeing where business students need more context, practice or confidence.

• **Editing:** more than ten years as a magazine editor, with an emphasis on clear, visual writing.

• **Research:** 63 course documents from 31 institutions, a 60-title catalogue, detailed textbook comparisons, and market and employer evidence.

• **Iteration:** a platform I can revise as tools change and students tell me what needs attention.

Ready now

• The full textbook, live as an interactive digital edition and in use at the University at Albany

• A Let's Build, an interactive lab and an auto-graded quiz in every chapter

• Student accounts, automatic quiz grading and an instructor dashboard

• Two running cases: the Grading Database and PetVax

• An Access route: Access labs and an Access appendix

Planned for the next edition

• The research-based 16-chapter outline below

• In-browser PostgreSQL (PGlite) practice workspaces inside the reader

• A planned MySQL route, to be tested on the shared cases

• Automated feedback on SQL and lab answers (prototype in testing)

• A small retail business, proposed as an alternative case for evaluation

Semester planning: fourteen teaching weeks is one possible route within a 14- to 16-week semester. The final workload, chapter selections and project schedule should be agreed with editors and adopting instructors.

Market research

How the market research was done

What we reviewed	NCES degree statistics, IPEDS completions through Data USA, published curriculum studies, and AACSB accreditation data. Syllabus, textbook and employer studies appear in their own sections below.
How	We retained published figures with their source years, separated degree counts from course and program evidence, and combined two of them into illustrative reach proxies. Each proxy is shown with its assumptions and would need publisher validation.
What we looked for	The scale of the student audience, where relevant courses are required, and which course settings are plausible adoption opportunities.

Finding 1: a large, required course

375,418 degrees, 2021-22	**Business bachelor's degrees awarded in the U.S.** About one in five of all bachelor's degrees.Source: NCES [1]
85% of 61 business schools	**Require at least one information systems course in the business core** 52 of 61 schools studied (89% of them AACSB-accredited); 3% required two IS courses.Source: Fustos, Morris and Haga [2]
12,092 awards in 2024	**Management information systems degrees awarded** Up 3.4% over the prior year. Counts all award levels reported, not only bachelor's degrees.Source: Data USA [3]
87.3% of 394 IS programs	**Offer a data and information management course** 344 of 394 U.S. undergraduate IS programs in business schools (data collected in 2013).Source: Hwang, Ma and Wang [4]
285 of 535 AACSB schools	**U.S. AACSB-accredited schools that offer an information systems program** A further 109 offer an undergraduate business analytics program (Table 2).Source: Zheng et al. [5]
82% of 74 IS master's programs	**Require a data, information and content management course** 61 of 74 programs at AACSB schools: a graduate audience for the same material.Source: Yang [6]
81,835 awards in 2024	**Accounting degrees awarded, an adjacent audience** Accounting information systems courses cover much of the same ground. All award levels reported.Source: Data USA [7]
1,098 institutions	**AACSB-accredited institutions worldwide** In 71 countries and territories, as of September 2026.Source: AACSB [8]

Award levels: the Data USA counts (MIS and accounting) are IPEDS completions across all award levels reported, not bachelor's degrees alone. The NCES figure counts bachelor's degrees only.

Two routes, one book

A commonality we found: **most business schools in the published studies require an information systems course of every business student, and most IS programs offer a database course.** The proposed routes address these two settings; fit must be reviewed against each course’s outcomes and available teaching time. In round numbers:

About 320,000 Illustrative reach proxy	**Business students who reach a required IS course** 375,418 business bachelor’s degrees (2021-22) × the 85% of business schools that require at least one IS course (2017 study).Source: Our estimate from [1], [2]

About 10,600 **Illustrative reach proxy**	**MIS students in programs with a database course** 12,092 MIS degrees across all award levels (2024) × the 87.3% of IS programs that offer the course (2015 study).Source: Our estimate from [3], [4]

Planning assumptions: each proxy multiplies a degree count by a historical, program-level share from a different source year. They are not enrollment-weighted course-seat estimates, and the MIS count covers all award levels. The two audiences overlap and must not be added together. Publisher enrollment and adoption data are needed to estimate addressable annual seats.

Adoption fit

Adoption fit is described qualitatively: which courses could use the courseware, what they need, and what a review should confirm. No revenue, seat or institution forecasts are made.

Segment	Likely need	Validation focus	Adoption opportunity
Primary Applied business database / MIS data management	Relational thinking, SQL, design, implementation, and business interpretation in one coherent learning path.	Course outcomes, annual sections and student seats, required database engine, incumbent materials, and switching criteria.	Full courseware adoption with readings, guided practice, independent labs, assessment, and instructor resources.
Business core data / IS course	Data meaning, information systems, queries, reporting, and a bounded project.	Which chapters fit the existing course and how much SQL and design it expects.	The business-core route through the same chapters.
Accounting information systems	Data meaning, controls, reporting and integrity.	Which modules fit the existing course.	Modular adoption or a companion pathway, subject to course fit.
Business analytics foundations	Data preparation, trustworthy SQL measures, and reporting.	Required design depth, analytical prerequisites, and reporting platform.	A selected chapter and lab route that supplies the database foundation for analytics.

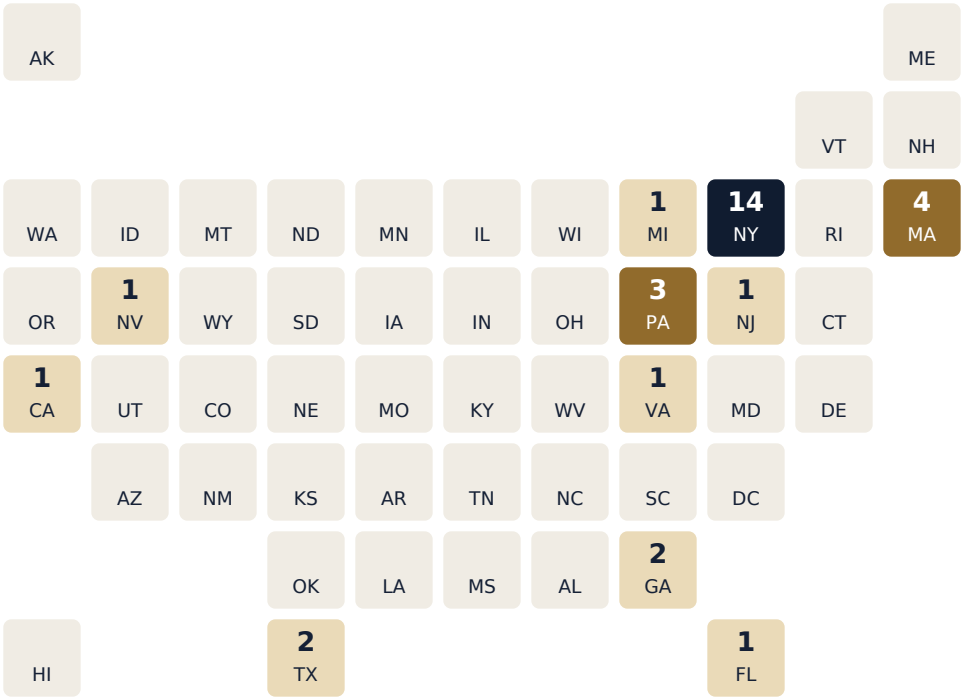
Main conclusion The evidence supports exploring two defined course routes: the applied business database / MIS course, which this book leads with, and the required business-core data / IS course. Their outcomes, tools and switching criteria should be validated separately.

Syllabus research

Why and how the syllabus research was done

Why	To check the outline against what courses across the country actually teach: which topics, which tools, which textbooks, how they assess, and what they say about AI.
What we collected	63 syllabi and course documents: syllabi, catalog descriptions and course schedules for database and business-data courses, mostly undergraduate, from the public websites of 31 universities. Retrieved October 1, 2026.
Why 63 documents but 31 institutions	Many institutions contribute more than one course, and some courses have more than one document. The 63 documents describe 54 distinct courses at 31 institutions. Of the 54 courses, 19 are business-database or business-core data courses, the closest match to this book.
How we read them	Archived documents and the earlier audits were reviewed for topics, platforms, textbooks, assessment and AI policies. Each set of findings retains its own sample and denominator. [S1]

The 31 institutions analyzed



14

New York. 8 SUNY campuses, plus Baruch, Brooklyn College, City Tech, NYU Stern, Syracuse and Hofstra.

12

East Coast. Rutgers, Temple, Penn State, Lafayette, Boston University, Northeastern, both UMass campuses, George Mason, Georgia, Georgia Tech and FIU.

5

National. UT Austin, UT Dallas, Michigan-Dearborn, San Jose State and UNLV.

D direct business-database course BC business-core data course. Unbadged courses are technical, follow-on or graduate comparisons. Together, the badged courses are the 19 closest comparators.

New York: SUNY8 institutions

• University at AlbanyCSI 508

• University at BuffaloDBCMGS 404, 351

• SUNY OswegoDISC 329

• SUNY CantonDBCCITA 215, 104

• SUNY Tompkins CortlandBCCAPS 131

• SUNY FarmingdaleBCS 260, 360, 390

• SUNY New PaltzCPS 593

• Stony Brook UniversityCSE 305

New York: CUNY and private6 institutions

• Baruch College, CUNYDCIS 3400, 4400

• NYU SternDTECH-UB 26

• Brooklyn College, CUNYBCCISC 1050

• Syracuse UniversityBCIST 195

• CUNY City TechCST 1204, 4724

• Hofstra UniversityCSC 056

Massachusetts4 institutions

• UMass DartmouthDMIS 332

• Boston UniversityCS 460, MET CS 669

• Northeastern UniversityCS 3200

• UMass AmherstCOMPSCI 345

Pennsylvania and New Jersey4 institutions

• Temple UniversityDMIS 2502

• Rutgers UniversityD33:136:470

• Penn StateIST 210

• Lafayette CollegeCS 320

South4 institutions

• George Mason University (VA)DMIS 310

• University of GeorgiaDMIST 4610

• Georgia TechDMGT 4058

• Florida InternationalCOP 4722

Midwest and West5 institutions

• UT Austin (TX)DMIS 325

• UT Dallas (TX)DITSS 4300

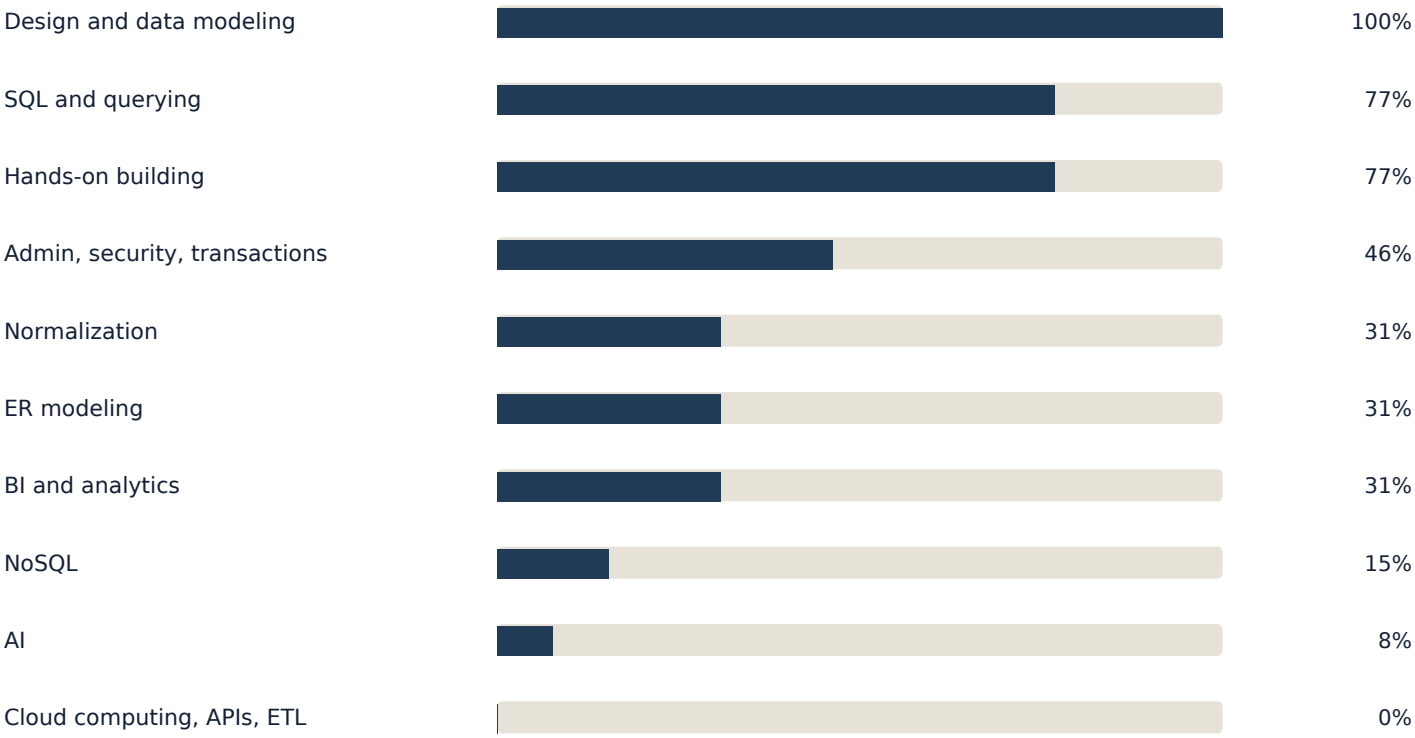
• Univ. of Michigan-DearbornDMIS 321

• San Jose State (CA)INFO 242

• UNLV (NV)MIS 766

What the courses teach, based on our analysis

Share of the 13 course descriptions in the earlier catalogue benchmark that mention each topic. These are lower bounds from course descriptions, not topic frequencies across the 63-document archive; a topic not documented in a description may still be taught.

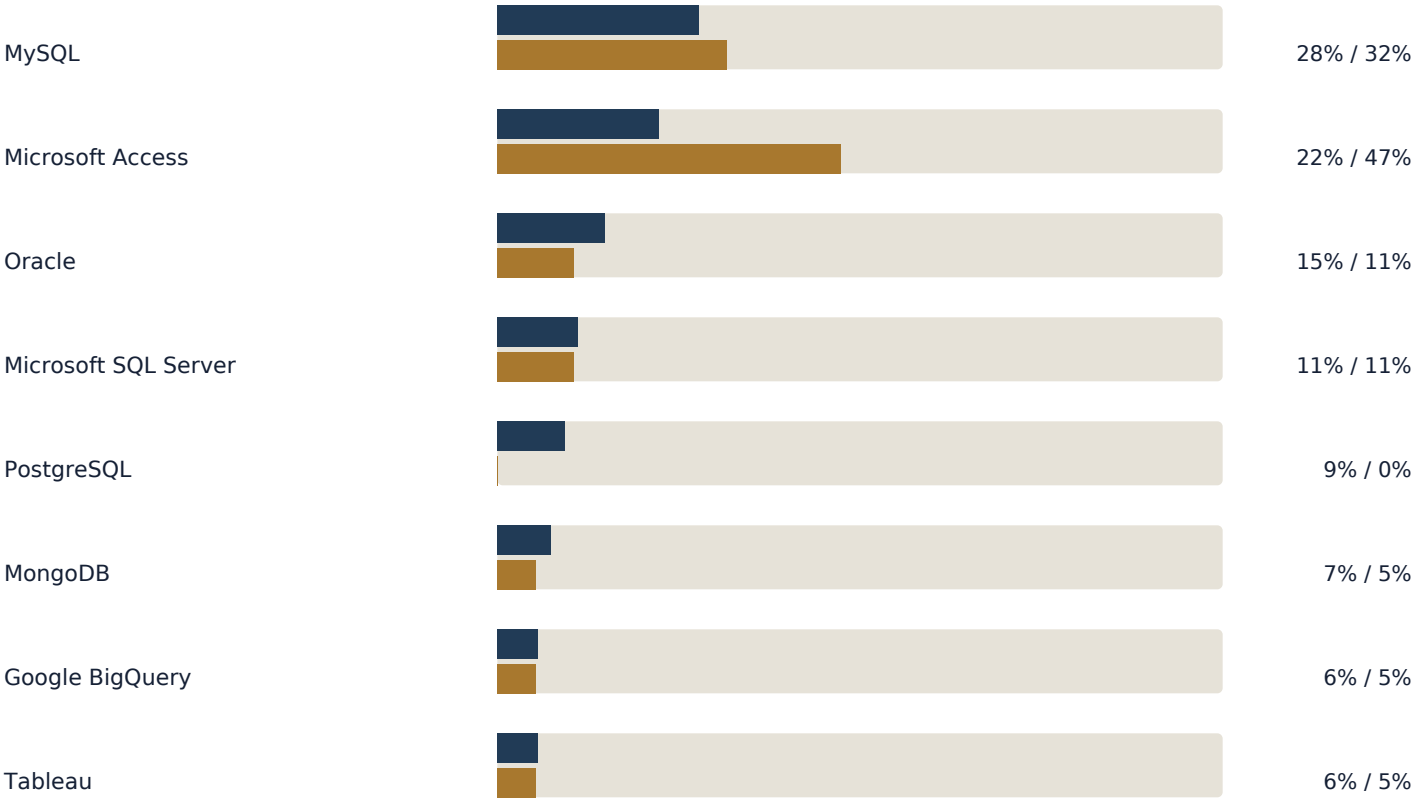


Which databases they use

Platform evidence is role-coded: **practical use** (the text assigns or demonstrates work with the tool), a **named option** , or a **software-list mention** only. The chart counts any of the three, as a share of all 54 course families and of the 19 direct and business-core families. Families can name more than one platform, and 20 of the 54 name none explicitly.

All 54 course families

19 direct and business-core families



Any mention (practical use, option or listed). Among the 19 closest comparators, Access has the largest footprint (9 of 19); across all 54 families, MySQL leads (15 of 54). Practical use alone is shown in the table.

Platform	Practical use	Option	Listed only	Course families	Institutions	Direct / business-core
MySQL	10	3	2	15	11	6
Microsoft Access	10	0	2	12	8	9
Oracle	4	2	2	8	7	2
Microsoft SQL Server	3	3	0	6	4	2
PostgreSQL	1	4	0	5	4	0
MongoDB	4	0	0	4	3	1
Google BigQuery	3	0	0	3	2	1

Platform	Practical use	Option	Listed only	Course families	Institutions	Direct / business-core
Tableau	3	0	0	3	2	1
SQLite	0	1	1	2	2	0

How they assess, and what they say about AI

Assessment

Staged and team projects recur across the documents reviewed, for example at SUNY Oswego, Baruch, Boston University, Rutgers and UT Dallas, usually alongside exams. The documents do not support a typical weighting across the archive.

AI

Only 12 of the 18 recent primary course offerings reviewed include a student AI-assistance provision. Five document no student rule; one points to a linked policy we could not inspect. These are 2024 to 2026 offerings, a subset of the full archive.

From evidence to design

A few examples show how specific courses informed the proposed learning path.

Institution	Course	Evidence date / level	Documented emphasis	Implication for this proposal
Rutgers	33:136:470	Fall 2026	SQL early, design/normalization, operational topics and a staged project.	Supports use-before-design and a cumulative project.
Baruch	CIS3400	Fall 2026 draft	Business requirements, ER modeling, normalization, SQL and an Access application.	Preserve application thinking and a substantive Access companion.
NYU Stern	TECH-UB26	Spring 2026	BigQuery SQL, cleaning, joins, aggregation, subqueries and window queries; ER concepts.	Supports browser/cloud SQL and an analytics-facing route.
Temple	MIS2502	Spring 2026	SQL, schemas, normalization, Python/JSON, data movement and analytics.	Connect query work to preparation and interpretation.
Georgia Tech	MGT4058	Fall 2026	Design, relational mapping, normalization, SQL and BI/warehouse concepts.	Protect the database core while connecting to reporting.

Institution	Course	Evidence date / level	Documented emphasis	Implication for this proposal
Boston University	MET-CS669	Spring 2026 · graduate	Design, SQL, operations and a project developed across six iterations.	Staged project structure is useful; graduate technical depth remains optional.
UT Dallas	ITSS4300	Spring 2026	SQL, modeling, normal forms, CTEs, windows, BI and security.	Provide advanced SQL extensions without removing essential design.
University of Georgia	MIST4610	Summer 2020	Modeling, MySQL, query work and visual answers to business questions.	Supports a query-to-communication chain; platform evidence is historical.

How did the findings shape the book?

Editorial interpretation: how each documented finding informed the proposed outline.

Design and data modeling Named in all 13 catalogue benchmark descriptions.	**A two-chapter design block:** requirements to ER diagrams, then normalization.Ch 7Ch 8
SQL and hands-on building Each appears explicitly in 10 of the 13 benchmark course descriptions (77%).	**Three SQL chapters,** a build-and-test chapter, and a Let's Build in every chapter.Ch 5Ch 6Ch 9Ch 10
Multiple platforms Access has the largest documented footprint in the Direct / Business-core subset; MySQL leads across all reviewed course families. The archive also includes Oracle, SQL Server, PostgreSQL and other systems.	**Teach SQL in the PostgreSQL dialect,** with a portable core and explicit dialect differences; keep the existing Access route, add a planned MySQL route to be tested on the same cases, and propose Supabase as the hosted PostgreSQL environment.App AApp BApp C
BI and analytics Named in 4 of 13 benchmark descriptions (31%).	**Analytics, reporting and Power BI chapters** close the loop from query to decision.Ch 12Ch 13Ch 14

AI and cloud AI / ML appears in 1 of 13 benchmark descriptions; cloud / APIs / ETL in none. This does not measure their prevalence across the full archive.	**An AI verification from Chapter 5 on,** and a dedicated chapter on big data, data warehouses and AI.Every chapterCh 15
Staged projects Staged, multi-phase projects recur across the documents reviewed.	**One staged project through the semester,** with graded checkpoints and an individual demonstration at the end.Studio ACh 11Ch 16

Method: 63 archived documents, 31 institutions, 54 grouped course families; topic percentages use a separate 13-course catalogue benchmark, and the AI-policy review uses 18 recent primary offerings. The searchable register below retains source dates and course roles.

Selected institution evidence

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Inspect all 63 course documents and their findings Expand the searchable course register

63 source documents

001 · SUNY Oswego · ISC329 Fall2012

Role: Direct. **Counting status:** Selected course family.

ER/UML, normalization, SQL, Access, web applications, transactions and warehousing; staged project.

Platform evidence: MySQL (used); Microsoft Access (used).

Original source

002 · University at Buffalo · MGS404 Fall2009

Role: Direct. **Counting status:** Selected course family.

SQL, ER/EER, normalization, physical design, performance, application development and data integration.

Platform evidence: MySQL (listed); Microsoft Access (listed); Oracle (listed).

Original source

003 · University at Buffalo · MGS351 Fall2026-as-labeled

Role: Business-core. **Counting status:** Selected course family.

Business IS with Access and Salesforce project work, modeling, queries, forms and reports.

Platform evidence: Microsoft Access (used); Salesforce (used).

Original source

004 · University at Buffalo · MGS613 Fall2012

Role: Graduate. **Counting status:** Selected course family.

ER, normalization, SQL/PLSQL, tuning, application development, warehouse/integration and distributed data.

Platform evidence: MySQL (listed); Microsoft Access (listed); Oracle (listed); SQLite (listed).

Original source

005 · SUNY Canton · CITA215 Master-Fall2025

Role: Direct. **Counting status:** Selected course family.

Relational structure, SQL, ER modeling, storage/indexing, transactions, recovery, cloud/distributed data and warehousing.

Platform evidence: No explicit practical platform established in reviewed text.

Original source

006 · SUNY Canton · CYBR216 Master-Fall2024

Role: Technical. **Counting status:** Selected course family.

SQL, ER modeling, normalization, integrity, security, backup, transactions, reporting, NoSQL and capstone.

Platform evidence: No explicit practical platform established in reviewed text.

Original source

007 · SUNY Canton · CITA104 Undated

Role: Business-core. **Counting status:** Selected course family.

Access database creation, maintenance, organization and queries; older software version.

Platform evidence: Microsoft Access (used).

Original source

008 · Stony Brook University · CSE305 Fall2026

Role: Technical. **Counting status:** Selected course family.

ER, normalization, relational algebra, SQL, physical indexing, optimization, triggers, NoSQL and working application project.

Platform evidence: No explicit practical platform established in reviewed text.

Original source

008a · Stony Brook University · CSE305-Schedule Fall2026

Role: Supplement. **Counting status:** supplement to 008.

Topic sequencing for the Fall 2026 CSE305 offering, including SQL, design and later system topics.

Platform evidence: See grouped/related course.

Original source

009 · Stony Brook University · CSE305-532 Spring2021

Role: Technical. **Counting status:** earlier version of 008.

CSE305/532 relational design, SQL, application project and deeper query-processing, indexing and optimization material.

Platform evidence: See grouped/related course.

Original source

010 · SUNY New Paltz · CPS593 Summer2020

Role: Graduate. **Counting status:** Selected course family.

Database fundamentals, SQL, applications, transactions, integrity, concurrency, security and distributed systems.

Platform evidence: MySQL (used).

Original source

011 · Baruch College · CIS3400 Fall2026-Draft

Role: Direct. **Counting status:** Selected course family.

Business requirements, ER modeling, relational mapping, normalization, SQL, Access application project, transactions and ethics.

Platform evidence: Microsoft Access (used).

Original source

012 · Baruch College · CIS4400 Fall2026-Draft

Role: Follow-on. **Counting status:** Selected course family.

Dimensional models, warehouse architecture, ETL/ELT, cloud, NoSQL and analytics project; assumes prior database competence.

Platform evidence: Google BigQuery (used); Tableau (used).

Original source

013 · NYU Stern · TECH-UB26 Spring2026

Role: Direct. **Counting status:** Selected course family.

SQL, cleaning, joins, aggregation, subqueries and window queries with relational/ER concepts; BigQuery.

Platform evidence: Google BigQuery (used).

Original source

014 · NYU Stern · INFO-GB2346 Spring2018

Role: Graduate. **Counting status:** Selected course family.

Python and SQL, APIs, collection, cleaning, storage and business insight project.

Platform evidence: No explicit practical platform established in reviewed text.

Original source

016 · Rutgers University · 33-136-470 Departmental

Role: Direct. **Counting status:** departmental duplicate; 030 is term-specific.

Relational/ER design, normalization, SQL, relational algebra, MySQL/PHP application project and optional NoSQL.

Platform evidence: See grouped/related course.

Original source

017 · Temple University · MIS2502 Spring2026

Role: Direct. **Counting status:** Selected course family.

SQL, schemas/keys, normalization and joins alongside Python, JSON, ETL, analytical storage, visualization and mining.

Platform evidence: MySQL (used).

Original source

018 · Boston University · MET-CS669 Spring2026

Role: Graduate. **Counting status:** Selected course family.

ER/EER/UML, normalization, SQL, stored procedures/triggers, transactions and a project developed over six iterations.

Platform evidence: Oracle (used); Microsoft SQL Server (option); PostgreSQL (option).

Original source

019 · Northeastern University · CS3200 Fall2026

Role: Technical. **Counting status:** Selected course family.

SQL-first sequence, ER/normalization, indexing, warehouse concepts, procedures/triggers, transactions, APIs and NoSQL.

Platform evidence: MySQL (used).

Original source

021 · George Mason University · MIS310 Master

Role: Direct. **Counting status:** Selected course family.

Database fundamentals, ER/relational design, normalization, SQL creation/population/querying and administration.

Platform evidence: Microsoft SQL Server (used).

Original source

022 · George Mason University · MIS410 Master

Role: Follow-on. **Counting status:** Selected course family.

Advanced design, SQL, procedures/triggers, indexes, security, reporting and application interfaces.

Platform evidence: Microsoft SQL Server (used).

Original source

023 · University of Georgia · MIST4610 Summer2020

Role: Direct. **Counting status:** Selected course family.

Relational modeling, normalization, MySQL SQL, analytical questions and business visualization.

Platform evidence: MySQL (used); Exploratory (used).

Original source

023a · University of Georgia · MIST4610-Assignments Summer2020

Role: Supplement. **Counting status:** supplement to 023.

Team data model, database implementation, diverse SQL queries and visual answers to business questions.

Platform evidence: See grouped/related course.

Original source

023b · University of Georgia · MIST4610-Grading Summer2020

Role: Supplement. **Counting status:** supplement to 023.

Assessment and grading structure for MIST4610.

Platform evidence: See grouped/related course.

Original source

023c · University of Georgia · MIST4610-Schedule Summer2020

Role: Supplement. **Counting status:** supplement to 023.

Relationships/modeling, joins, normalization, advanced SQL and business analytics sequence.

Platform evidence: See grouped/related course.

Original source

024 · UT Austin · MIS325-BAX327 Spring2024

Role: Direct. **Counting status:** Selected course family.

ER design, logical mapping/normalization, DDL/DML, joins/aggregation/subqueries, PL/SQL, security, warehouse/ETL and NoSQL.

Platform evidence: Oracle (used); MongoDB (used); Tableau (used).

Original source

025 · UT Dallas · MIS6326 Spring2022

Role: Graduate. **Counting status:** earlier version of 054.

Design, SQL, integration, warehousing, quality/governance, transactions and performance.

Platform evidence: See grouped/related course.

Original source

026 · UT Dallas · BUAN6v99-503 Fall2025

Role: Graduate. **Counting status:** Selected course family.

Python/ORM applications, tuning, NoSQL, distributed systems, embeddings, vector retrieval, RAG and AI-assisted querying.

Platform evidence: SQLite (option); PostgreSQL (option); MySQL (option); MongoDB (used); Redis (used).

Original source

027 · San Jose State University · INFO242 Spring2022

Role: Graduate. **Counting status:** Selected course family.

ER design, normalization, Oracle SQL, administration, concurrency, web applications, privacy and staged case project.

Platform evidence: No explicit practical platform established in reviewed text.

Original source

028 · Florida International University · COP4722 Fall2020

Role: Technical. **Counting status:** Selected course family.

Query processing/optimization, transactions, concurrency, web/XML, warehousing and specialized database types.

Platform evidence: No explicit practical platform established in reviewed text.

Original source

029 · Florida International University · COP4751 Spring2021

Role: Follow-on. **Counting status:** Selected course family.

Advanced database implementation, transactions, web data, analytics and nonrelational/big-data topics.

Platform evidence: No explicit practical platform established in reviewed text.

Original source

030 · Rutgers University · 33-136-470 Fall2026

Role: Direct. **Counting status:** Selected course family.

Early SQL through joins/views; then design/normalization, indexes, transactions, security and architectures; two-phase data project.

Platform evidence: MySQL (used).

Original source: archived copy held by the author; no public URL.

031 · Rutgers University · 22-544-645_26-198-641 Fall2026

Role: Graduate. **Counting status:** Selected course family.

Assumes SQL/design; cloud storage, multiple NoSQL models, distributed systems, Spark/streaming and data engineering projects.

Platform evidence: MongoDB (used); Neo4j (used); Amazon Web Services (used).

Original source: archived copy held by the author; no public URL.

032 · Lafayette College · CS320 Fall2020

Role: Technical. **Counting status:** Selected course family.

Relational model/algebra, SQL, ER, constraints/triggers, transactions, distributed/NoSQL and client-oriented application project.

Platform evidence: No explicit practical platform established in reviewed text.

Original source: archived copy held by the author; no public URL.

033 · Georgia Tech · MGT4058 Fall2026

Role: Direct. **Counting status:** Selected course family.

ER design, relational mapping, normalization, SQL, implementation, BI/warehousing and awareness of AI-enabled support.

Platform evidence: No explicit practical platform established in reviewed text.

Original source

034 · Boston University · CS460 Fall2026

Role: Technical. **Counting status:** Selected course family.

ER, relational algebra/SQL, storage, query processing, transactions, concurrency, replication and NoSQL/MapReduce.

Platform evidence: No explicit practical platform established in reviewed text.

Original source

036 · University at Albany · CSI508 Spring2026

Role: Graduate. **Counting status:** Selected course family.

Relational algebra, SQL, schema design/normalization, indexing, query optimization, transactions and concurrency.

Platform evidence: No explicit practical platform established in reviewed text.

Original source

037 · UNLV · MIS766 SummerII2020-Sample

Role: Graduate. **Counting status:** Selected course family.

Managerial cases, relational design, SQL, transactions, quality, governance/privacy, NoSQL and organizational project.

Platform evidence: No explicit practical platform established in reviewed text.

Original source

038 · Michigan Dearborn · MIS321 Winter2007

Role: Direct. **Counting status:** Selected course family.

Conceptual ER design, logical normalization, physical design/indexes and Access/SQL team implementation.

Platform evidence: Microsoft Access (used).

Original source

039 · Michigan Dearborn · MIS322 Winter2007

Role: Follow-on. **Counting status:** Selected course family.

Oracle SQL, functions/subqueries, schema operations, privileges, concurrency and database programming; prior design assumed.

Platform evidence: Oracle (used).

Original source

041 · UT Dallas · ITSS4300 Fall2016

Role: Direct. **Counting status:** earlier version of 055.

ER/normalization/SQL, performance, concurrency/security, BI/big data and project work.

Platform evidence: See grouped/related course.

Original source

042 · UT Dallas · ITSS4380 Spring2025

Role: Follow-on. **Counting status:** Selected course family.

Advanced SQL, stored procedures/triggers, NoSQL/MongoDB, administration and a multi-phase project; prerequisite ITSS4300.

Platform evidence: MySQL (used); MongoDB (used).

Original source

043 · Hofstra University · CSC056 Fall2005

Role: Technical. **Counting status:** Selected course family.

SQL, ER/normalization, algebra/calculus, constraints/triggers, web/XML and DBMS implementation topics.

Platform evidence: No explicit practical platform established in reviewed text.

Original source

044 · UMass Amherst · COMPSCI345 Undated

Role: Technical. **Counting status:** Selected course family.

SQL-first queries and updates, then ER/normalization, PostgreSQL indexing/views and transactions.

Platform evidence: PostgreSQL (used).

Original source

045 · UMass Amherst · CS197Q Undated

Role: Technical. **Counting status:** Selected course family.

Relational/ER concepts, SQL retrieval, DDL/CRUD, integrity/security and MySQL project.

Platform evidence: MySQL (used).

Original source

046 · Baruch College · CIS4130 Fall2024

Role: Follow-on. **Counting status:** Selected course family.

GCP, Hadoop/Spark/PySpark, machine-learning pipelines, evaluation and streaming.

Platform evidence: No explicit practical platform established in reviewed text.

Original source

047 · Baruch College · CIS9340 Spring2025-Draft

Role: Graduate. **Counting status:** Selected course family.

ER design, normalization, SQL/algebra, Access implementation, transactions, security and ethics.

Platform evidence: Microsoft Access (used).

Original source

048 · Baruch College · CIS9440 Undated-Draft

Role: Graduate. **Counting status:** Selected course family.

Dimensional models, ETL/ELT, cloud platforms, NoSQL/big-data architectures and analytics project.

Platform evidence: Google BigQuery (used); Tableau (used).

Original source

049 · Baruch College · CIS9760 Spring2025

Role: Graduate. **Counting status:** Selected course family.

Cloud, Hadoop/Spark, machine-learning data pipelines and streaming.

Platform evidence: No explicit practical platform established in reviewed text.

Original source

050 · Penn State · IST210 Spring2009

Role: Technical. **Counting status:** Selected course family.

Conceptual design, refinement, implementation, SQL, administration and project work.

Platform evidence: Microsoft SQL Server (used); MySQL (option); PostgreSQL (option); Oracle (option); IBM Db2 (option).

Original source

053 · UMass Dartmouth · MIS332 Spring2026-Draft

Role: Direct. **Counting status:** Selected course family.

Relational databases, complex SQL, design, storage, transactions, security, programming and NoSQL.

Platform evidence: MySQL (used); Microsoft Access (used).

Original source

054 · UT Dallas · MIS6326 Fall2026

Role: Graduate. **Counting status:** Selected course family.

Relational/ER modeling, normalization, several weeks of SQL, views and transactions; multiple DBMS options.

Platform evidence: Microsoft SQL Server (option); MySQL (option); PostgreSQL (option); Oracle (option).

Original source

055 · UT Dallas · ITSS4300 Spring2026

Role: Direct. **Counting status:** Selected course family.

Early SQL and modeling, 1NF-3NF, joins/unions, CTEs, window functions, BI, tuning, DDL/views and security.

Platform evidence: Microsoft SQL Server (option); Tableau Prep (used).

Original source

056 · SUNY Tompkins Cortland · CAPS131 Master2025-26

Role: Business-core. **Counting status:** Selected course family.

Access tables, queries, forms and reports in a one-credit course.

Platform evidence: Microsoft Access (used).

Original source

057 · CUNY City Tech · CST1204 Master

Role: Technical. **Counting status:** Selected course family.

Relational design, normalization, keys/constraints, SQL DDL/DML, joins/subqueries, views, administration and import/export.

Platform evidence: Oracle (used).

Original source

058 · CUNY City Tech · CST4724 Undated

Role: Follow-on. **Counting status:** Selected course family.

Key-value, document, column-oriented and graph databases, queries and ethical issues.

Platform evidence: No explicit practical platform established in reviewed text.

Original source

059 · Brooklyn College · CISC1050 Spring2026

Role: Business-core. **Counting status:** Selected course family.

Broad computer literacy, Office applications, spreadsheets and a short Access unit.

Platform evidence: Microsoft Access (used).

Original source

060 · University at Buffalo · LIS569 Spring2012

Role: Graduate. **Counting status:** Selected course family.

ER design, normalization, SQL, views, interface/report integration and PHP/MySQL/Access project.

Platform evidence: MySQL (used); Microsoft Access (used).

Original source

062 · CUNY City Tech · CST4724 Undated

Role: Follow-on. **Counting status:** duplicate representation of 058.

CST4724 NoSQL scope and outcomes in abbreviated PDF form.

Platform evidence: See grouped/related course.

Original source

063 · SUNY Farmingdale · BCS260 Undated

Role: Technical. **Counting status:** Selected course family.

ER design, normalization through BCNF, SQL, development lifecycle and a documented database capstone.

Platform evidence: No explicit practical platform established in reviewed text.

Original source

064 · SUNY Farmingdale · BCS360 Undated

Role: Follow-on. **Counting status:** Selected course family.

Advanced SQL/PLSQL, procedures/cursors/triggers, transactions, recovery, higher normalization and security.

Platform evidence: No explicit practical platform established in reviewed text.

Original source

065 · SUNY Farmingdale · BCS390 Undated

Role: Follow-on. **Counting status:** Selected course family.

DBMS installation/configuration, users/privileges, storage, backup/restore, performance and integrity.

Platform evidence: No explicit practical platform established in reviewed text.

Original source

066 · Syracuse University · IST195 Fall2023

Role: Business-core. **Counting status:** Selected course family.

Broad hardware, operating systems, networks, cloud, ethics/AI, spreadsheets/visualization and a brief database unit.

Platform evidence: No explicit practical platform established in reviewed text.

Original source

Search filters document records, including grouped supplements. The aggregate denominator remains 54 course families.

Labor, employer and industry research

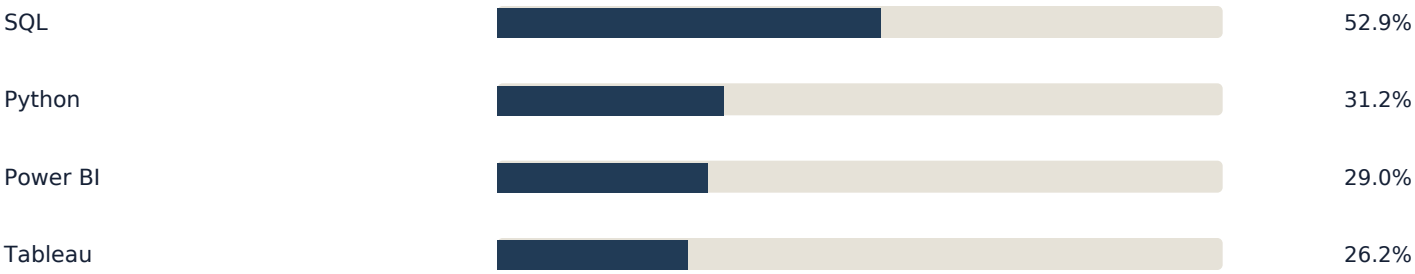
What we researched

A 2026 analysis by 365 Data Science of 1,000 Indeed USA data analyst postings (855 relevant), U.S. Bureau of Labor Statistics projections for 2025-35, and the World Economic Forum employer survey.

What employers ask for

We looked at what employers ask for in the skills this book teaches.

52.9% of job postings	**U.S. data analyst job postings that ask for SQL** Of 855 relevant postings in a vendor analysis of 1,000 Indeed USA listings. Python appears in 31.2%, Power BI in 29.0% and Tableau in 26.2%.Source: 365 Data Science [9]
+35% growth, 2025 to 2035	**Projected growth in data scientist jobs** Against 3% for all occupations. Median pay: $120,230 a year (May 2025).Source: U.S. Bureau of Labor Statistics [10]
+9% growth, 2025 to 2035	**Projected growth in database architect jobs** Median pay: $139,500 a year (May 2025).Source: U.S. Bureau of Labor Statistics [11]
About two-thirds of employers worldwide	**Plan to hire for AI skills** AI and big data are the fastest-growing skills employers report.Source: World Economic Forum [12]



Share of 855 U.S. data analyst job postings that list each skill.

Evidence	Finding	Courseware implication
BLS: data scientists, 2025-35	**35% projected growth** , compared with 3% for all occupations; about **24,800 openings annually** . [L1]	Connect data preparation and queries to interpretation, communication, and decisions.
BLS: database architects / administrators, 2025-35	Architects: **9%** growth. Administrators: **0%** . Combined: about **7,300 openings annually** . [L2]	Retain modeling, quality, permissions, backup, and system judgment alongside analytics.
WEF: Future of Jobs Report 2025	Survey of **1,000+ employers** , representing **14 million+ workers** across **55 economies** ; roughly **7 in 10** identify analytical thinking as essential. [L3]	Require students to explain a result, test its reliability, and justify an action.
WEF: skills outlook to 2030	AI and big data lead the skills employers expect to grow fastest. [L3]	Integrate responsible AI use, verification, data provenance, and limits across the learning experience.

Main conclusion SQL, reporting and judgment belong together. Students need to produce a result, establish that it is reliable, and communicate what an organization should do with it.

Job-posting shares describe this sample of 855 postings, not all U.S. hiring. BLS growth projections (the change in employment, 2025-35) and annual openings are separate measures: openings include replacement hiring, so they can be substantial even where growth is flat. Global employer expectations (WEF) are distinct from U.S. job-posting requirements.

Textbook research

How the textbook research was done

I compared the books already used in these courses, starting with four especially relevant titles. The wider audit records 60 named books and course texts. I examined their chapter sequences, practice environments and course fit, then asked what this courseware should keep and where it could add value.

What the course collection tells us about textbook choice

60 titles	**Textbooks named in the syllabi we reviewed** No title appeared at more than 3 of the 31 schools reviewed. Instructor-made materials also appear. This describes the reviewed collection, not national market share.Source: Syllabus research, below
2021 last revision	**Cengage's applied business-database title** Concepts of Database Management, 10th edition, is built on Access 2019 and SQL Server 2019. It has served the course well, and the tools and topics have moved on since.Source: Cengage [13]

The textbooks reviewed

Database Systems: Design, Implementation, and Management

Coronel and Morris · Cengage, 15th edition (©2027)

Chapters	16 in 6 parts
SQL starts	Chapter 4, before ER modeling
Tools	Auto-graded database labs in MindTap
Best fit	IS and MIS majors who need depth

Concepts of Database Management

Friedrichsen et al. · Cengage, 10th edition (2021)

Chapters	9
SQL starts	Chapters 2 to 3
Tools	Access 2019, SQL Server 2019; MindTap and SAM
Best fit	Applied business courses

Database Concepts

Kroenke, Auer, Vandenberg and Yoder · Pearson, 10th edition

Chapters	7, with more online
SQL starts	Chapter 3
Tools	Access, MySQL 8.0, Azure; running projects
Best fit	Short, Access-first courses

Modern Database Management

Hoffer, Topi, Venkataraman and Bala · Pearson, 14th edition

Chapters	13
SQL starts	Chapter 5
Tools	Oracle, SQL Server, MySQL; an analytics chapter and State of AI sections
Best fit	Managerial MIS courses

Database Systems with SQL and zyLabs

Winsberg et al. · zyBooks

Chapters	12 interactive chapters

SQL starts	Early
Tools	Built-in SQL sandbox; MySQL and MySQL Workbench chapters
Best fit	Practice-heavy courses

Murach's MySQL

Murach · 4th edition

Chapters	Query-first workbook
SQL starts	Chapter 1
Tools	MySQL Workbench, AWS
Best fit	SQL skills courses

Using Data to Drive Business Performance

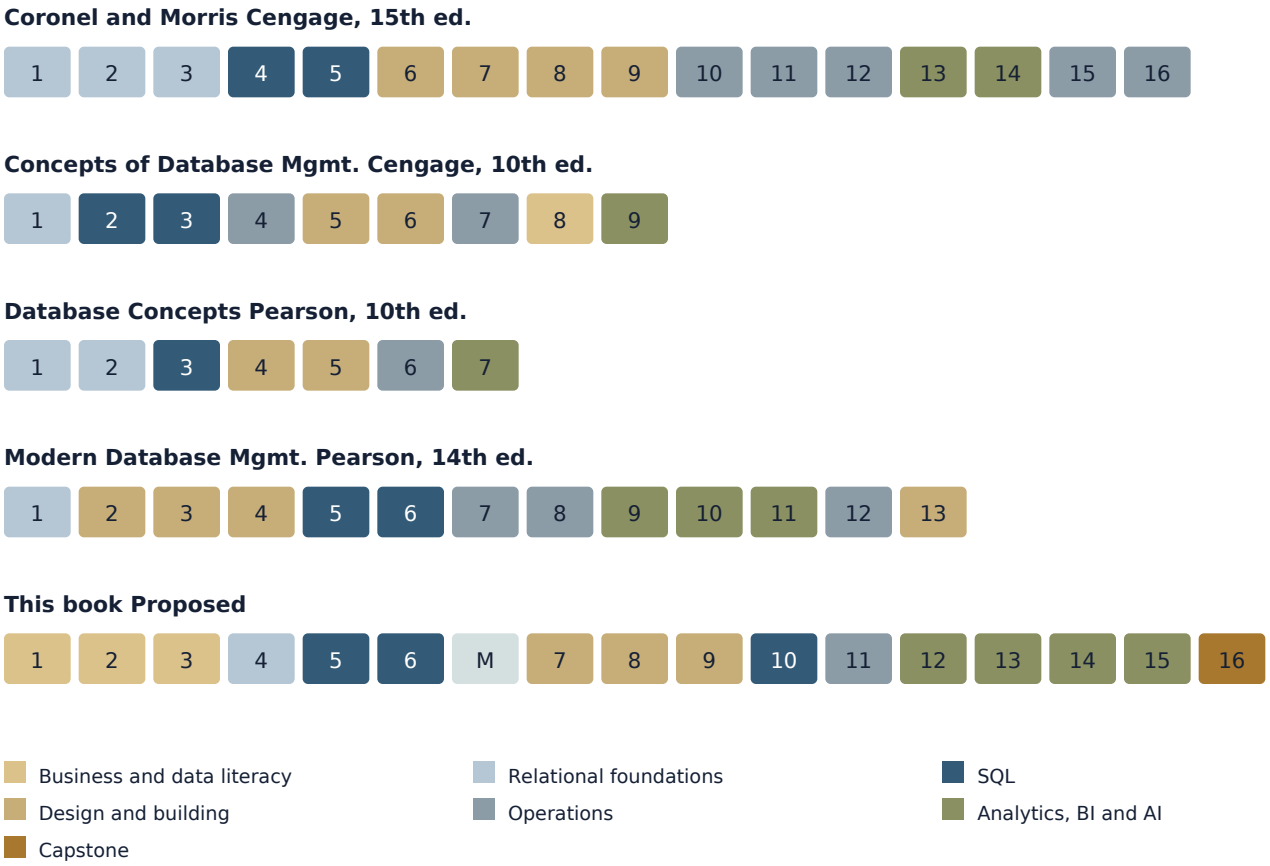
Dvir · proposed next edition

Chapters	16 in 5 parts
SQL starts	Chapter 5, after data literacy
Tools	PostgreSQL and Supabase; MySQL and Access routes; Power BI
Best fit	Business-core and MIS courses

Publisher pages: Coronel and Morris 15e; Concepts of Database Management 10e; Database Concepts 10e; Modern Database Management 14e.

Chapter by chapter, in order

We coded each chapter of the four main textbooks by its main topic to see which topics the books have in common and in what order they teach them. Each square is one chapter. The last row is the proposed outline of this book.



M marks the midterm Integration Studio. Topic classification is ours, from publisher tables of contents.

What it tells us: the four compared sequences share relational foundations, SQL and design. The proposed edition keeps that core and extends its opening business / data-literacy sequence into dedicated analysis, reporting, Power BI and capstone work.

What the leading books do well, and this book keeps

- **The relational core:** keys, relationships, ER modeling and normalization to third normal form
- **Early, hands-on SQL,** as in Coronel 15th edition, Kroenke and zyBooks
- **Administration, transactions and security** as a full chapter
- **Warehousing, star schemas and NoSQL** as the bridge to analytics

Where they stop, and this book continues

- **The four compared chapter sequences open with database foundations rather than a three-chapter business-performance, data-literacy and information-organization sequence.** This book does, in Chapters 1 to 3.

• **A dedicated Power BI studio is a proposed distinction in this comparison,** alongside analytics, visual reporting and a capstone decision memo. Current competing texts already cover aspects of BI and analytics.

• **Competing texts cover AI as a topic;** Modern Database Management, for example, adds State of AI sections. This design makes checking AI-assisted work a recurring practice, with structured verification activities from Chapter 5 onward.

• **Courses and books support different database products.** This proposal keeps PostgreSQL central, keeps the existing Access route, and plans a MySQL route to be tested on the same cases.

Inspect the four detailed book outlines

Database Systems: Design, Implementation, & Management Cengage

Coronel and Morris · 15th edition; 14th retained as historical baseline [P2]

The existing course proposition

Broad database design, implementation and systems instruction. Its digital resources already include SQL practice; the current edition places SQL before formal design.

The proposed distinction

The proposed book narrows the primary audience to applied business learners and makes validation, reporting and a decision memo part of one repeated workflow.

Chapter-level sequence

• Database systems

• Data models

• Relational foundations

• SQL introduction

• Advanced SQL

• ER modeling

• Advanced modeling

• Normalization

• Database design

• Transactions and concurrency

• Query performance

- Distributed databases
- BI and warehouses
- Big data and NoSQL
- Connectivity and web technologies
- Administration and security

The 15th-edition sequence is paraphrased from the earlier contents audit; the official current listing was checked for edition and scope. The uploaded 14th edition instead puts modeling/normalization before SQL. Do not use its sequence to describe the 15th.

Concepts of Database Management Cengage

Friedrichsen, Ruffolo, Monk, Starks, Pratt and Last · 10th edition [P3]

The existing course proposition

A concise applied introduction with QBE/algebra, SQL, normalization, relationships, operations, careers and trends. Includes cases and Access training in MindTap.

The proposed distinction

The nearest portfolio conversation: a business-performance framing with verified analytical SQL, a proposed browser-to-hosted PostgreSQL path, BI work and an integrated decision deliverable. This is a potential position, not an agreed replacement.

Chapter-level sequence

- Introduction to database management
- Relational model, QBE and algebra
- Relational SQL
- Advanced relational topics
- Normalization
- Relationships in database design
- DBMS processes and services
- Database careers
- Industry trends

The official nine-module sequence is paraphrased. Appendices address a complete design example, SQL, FAQs, MySQL and requirements analysis.

Database Concepts Pearson

Kroenke, Auer, Vandenberg and Yoder · 10th edition; uploaded PDF is 9th [P4]

The existing course proposition

A compact hands-on database text with multiple running examples/projects, software flexibility, cloud coverage and MySQL extensions.

The proposed distinction

The proposed distinction is the sustained business-measure-to-decision chain and explicit evidence requirements, rather than the mere existence of business cases, cloud topics or early SQL.

Chapter-level sequence

- Getting started
- Relational foundations
- SQL
- ER data models
- Database design
- Administration and cloud
- BI, warehouses and big data

Seven-part chapter sequence paraphrased using publisher/reference-edition audit. The older uploaded PDF supports historical analysis and must not be labeled the current edition.

Modern Database Management Pearson

Hoffer, Topi, Venkataraman and Bala · 14th edition [P5]

The existing course proposition

A comprehensive organizational database text. Current coverage extends from data to analysis, with an analytics chapter, notes on cloud-based solutions and new State of AI sections.

The proposed distinction

A serious comparison: the proposal must demonstrate a more bounded business-course workload and a coherent guided-to-independent learning experience; it cannot claim competitors omit analytics or AI.

Chapter-level sequence

- Database environment and development
- Organizational data modeling
- Enhanced ER models
- Logical design and relational foundations
- SQL introduction
- Advanced SQL
- Databases in applications
- Physical design and infrastructure
- Big data
- Integrated platforms and warehouses
- Data-to-insight analytics
- Administration and data quality
- Object-oriented modeling

Current official chapter-level sequence paraphrased. Advanced normal forms and notation appear in appendices.

Explore the full 60-title catalogue Search by book, author, publisher, or institution

60 named titles in the prior audit

Database Management Systems McGraw Hill · 3 institutions

Raghu Ramakrishnan; Johannes Gehrke

2 required/main; 1 optional-only; 0 listing-only institutions.

Hofstra University; Rutgers University; Stony Brook University

Course	Term	Role	Edition as cited
Stony Brook University · CSE305-532	Spring2021	Required	3rd
Rutgers University · 33-136-470	Departmental	Suggested, not required	3rd

Course	Term	Role	Edition as cited
Hofstra University · CSC056	Fall2005	Required	3rd

Database Systems: The Complete Book Pearson · 3 institutions

Hector Garcia-Molina; Jeffrey Ullman; Jennifer Widom

2 required/main; 1 optional-only; 0 listing-only institutions.

Boston University; Lafayette College; UMass Amherst

Course	Term	Role	Edition as cited
Lafayette College · CS320	Fall2020	Required	2nd
Boston University · CS460	Fall2026	Optional	2nd
UMass Amherst · COMPSCI345	Undated	Listed main text	2nd

Fundamentals of Database Systems Pearson · 3 institutions

Ramez Elmasri; Shamkant Navathe

1 required/main; 2 optional-only; 0 listing-only institutions.

Florida International University; Michigan Dearborn; UT Dallas

Course	Term	Role	Edition as cited
Florida International University · COP4722	Fall2020	Listed main text	7th
Florida International University · COP4751	Spring2021	Listed main text	7th
Michigan Dearborn · MIS321	Winter2007	Recommended reference	5th
Michigan Dearborn · MIS322	Winter2007	Recommended reference	5th
UT Dallas · ITSS4380	Spring2025	Suggested	7th

Principles of Database Management: The Practical Guide to Storing, Managing and Analyzing Big and Small Data Cambridge University Press · 3 institutions

Wilfried Lemahieu; Seppe vanden Broucke; Bart Baesens

1 required/main; 2 optional-only; 0 listing-only institutions.

Northeastern University; Stony Brook University; UNLV

Course	Term	Role	Edition as cited
Stony Brook University · CSE305	Fall2026	Optional NoSQL reference	2018
Northeastern University · CS3200	Fall2026	Optional	2018
UNLV · MIS766	SummerII2020-Sample	Required	2018

A Guide to SQL Cengage · 2 institutions

Philip Pratt; Mary Last et al.

1 required/main; 1 optional-only; 0 listing-only institutions.

CUNY City Tech; University at Buffalo

Course	Term	Role	Edition as cited
University at Buffalo · MGS404	Fall2009	Optional	7th
CUNY City Tech · CST1204	Master	Required	9th as cited; ISBN inconsistent

City Tech lists 9th edition with ISBN 9781111527273 and year 2009. Those details conflict; the title counts, but the edition is not treated as resolved.

Database System Concepts McGraw Hill · 2 institutions

Abraham Silberschatz; Henry Korth; S. Sudarshan

1 required/main; 1 optional-only; 0 listing-only institutions.

SUNY Canton; University at Albany

Course	Term	Role	Edition as cited
SUNY Canton · CYBR216	Master-Fall2024	Suggested	Not stated
University at Albany · CSI508	Spring2026	Listed main text	7th

Database Systems with SQL and zyLabs Wiley / zyBooks · 2 institutions

Paul Winsberg et al.

2 required/main; 0 optional-only; 0 listing-only institutions.

Rutgers University; UMass Dartmouth

Course	Term	Role	Edition as cited
Rutgers University · 33-136-470	Fall2026	Purchase directed / main text	2025
UMass Dartmouth · MIS332	Spring2026-Draft	Listed course text	Not stated

Two schools explicitly name this title. SUNY Canton names a customized zyBooks course text separately; it is not merged into this title count.

Database Systems: A Practical Approach to Design, Implementation, and Management Pearson · 2 institutions

Thomas Connolly; Carolyn Begg

2 required/main; 0 optional-only; 0 listing-only institutions.

Baruch College; SUNY Oswego

Course	Term	Role	Edition as cited
SUNY Oswego · ISC329	Fall2012	Listed main text	5th
Baruch College · CIS9340	Spring2025-Draft	Main text / assigned reading	6th; 5th accepted

Database Systems: Design, Implementation, & Management Cengage · 2 institutions

Carlos Coronel; Steven Morris

2 required/main; 0 optional-only; 0 listing-only institutions.

Boston University; UT Dallas

Course	Term	Role	Edition as cited
Boston University · MET-CS669	Spring2026	Required	14th; 13th accepted
UT Dallas · MIS6326	Spring2022	Listed main text	13th
UT Dallas · ITSS4300	Fall2016	Required	12th
UT Dallas · ITSS4380	Spring2025	Suggested	13th
UT Dallas · ITSS4300	Spring2026	Suggested	14th

Database Systems: Introduction to Databases and Data Warehouses Prospect Press · 2 institutions

Nenad Jukic; Susan Vrbsky; Svetlozar Nestorov; Abhishek Sharma

0 required/main; 2 optional-only; 0 listing-only institutions.

Georgia Tech; NYU Stern

Course	Term	Role	Edition as cited
NYU Stern · TECH-UB26	Spring2026	Optional	Not stated
Georgia Tech · MGT4058	Fall2026	Optional, recommended	2nd

Modern Database Management Pearson · 2 institutions

Hoffer et al.

2 required/main; 0 optional-only; 0 listing-only institutions.

Michigan Dearborn; University at Buffalo

Course	Term	Role	Edition as cited
University at Buffalo · MGS404	Fall2009	Required	9th
University at Buffalo · MGS613	Fall2012	Required	11th
Michigan Dearborn · MIS321	Winter2007	Required	8th
Michigan Dearborn · MIS322	Winter2007	Required	8th
University at Buffalo · LIS569	Spring2012	Required	10th

Murach’s MySQL Mike Murach & Associates · 2 institutions

Joel Murach

0 required/main; 2 optional-only; 0 listing-only institutions.

Northeastern University; UT Dallas

Course	Term	Role	Edition as cited
Northeastern University · CS3200	Fall2026	Strongly recommended	4th; 3rd accepted
UT Dallas · ITSS4300	Fall2016	Optional	2nd
UT Dallas · ITSS4380	Spring2025	Suggested	4th

Building the Data Warehouse Wiley / zyBooks · 1 institution

W. H. Inmon

0 required/main; 1 optional-only; 0 listing-only institutions.

Baruch College

Course	Term	Role	Edition as cited
Baruch College · CIS4400	Fall2026-Draft	Recommended	4th

Course	Term	Role	Edition as cited
Baruch College · CIS9440	Undated-Draft	Recommended	4th

Business Intelligence Guidebook: From Data Integration to Analytics Elsevier / Morgan Kaufmann · 1 institution

Rick Sherman

1 required/main; 0 optional-only; 0 listing-only institutions.

Baruch College

Course	Term	Role	Edition as cited
Baruch College · CIS4400	Fall2026-Draft	Main text / assigned reading	2014
Baruch College · CIS9440	Undated-Draft	Main text / assigned reading	2014

CITA 215: Database Application and Concepts (custom zyBooks) Wiley / zyBooks · 1 institution

Not identified in syllabus

1 required/main; 0 optional-only; 0 listing-only institutions.

SUNY Canton

Course	Term	Role	Edition as cited
SUNY Canton · CITA215	Master-Fall2025	Listed course text	ISBN 9798203387394

Custom course title. Publisher catalog outline can establish the product family, not the exact selected chapters in the custom course.

Cassandra: The Definitive Guide O’Reilly Media · 1 institution

Jeff Carpenter; Eben Hewitt

1 required/main; 0 optional-only; 0 listing-only institutions.

Rutgers University

Course	Term	Role	Edition as cited
Rutgers University · 22-544-645_26-198-641	Fall2026	Required	Revised 3rd, 2022

Computer Applications for Managers Lumen Learning / OER · 1 institution

Lumen Learning

0 required/main; 0 optional-only; 1 listing-only institutions.

Brooklyn College

Course	Term	Role	Edition as cited
Brooklyn College · CISC1050	Spring2026	Free course material	Not stated

Computing Essentials 2023: Making IT Work for You McGraw Hill · 1 institution

Timothy; Linda; Daniel O’Leary

1 required/main; 0 optional-only; 0 listing-only institutions.

Syracuse University

Course	Term	Role	Edition as cited
Syracuse University · IST195	Fall2023	Required	2023

Concepts of Database Management Cengage · 1 institution

Friedrichsen; Ruffolo; Monk; Starks; Pratt; Last

0 required/main; 1 optional-only; 0 listing-only institutions.

Baruch College

Course	Term	Role	Edition as cited
Baruch College · CIS3400	Fall2026-Draft	Optional	8th, 9th or 10th

Data Engineering with Google Cloud Platform Packt · 1 institution

Adi Wijaya

0 required/main; 1 optional-only; 0 listing-only institutions.

Baruch College

Course	Term	Role	Edition as cited
Baruch College · CIS4130	Fall2024	Optional	2022
Baruch College · CIS9760	Spring2025	Optional	2022

Data Management: Databases and Organizations eGreen Press · 1 institution

Richard T. Watson

1 required/main; 0 optional-only; 0 listing-only institutions.

University of Georgia

Course	Term	Role	Edition as cited
University of Georgia · MIST4610	Summer2020	Listed main text	2017

Data Science on the Google Cloud Platform O’Reilly Media · 1 institution

Valliappa Lakshmanan

0 required/main; 1 optional-only; 0 listing-only institutions.

Baruch College

Course	Term	Role	Edition as cited
Baruch College · CIS4130	Fall2024	Optional	2nd
Baruch College · CIS9760	Spring2025	Optional	2nd

Data-Intensive Text Processing with MapReduce Morgan & Claypool / Springer Nature · 1 institution

Jimmy Lin; Chris Dyer

0 required/main; 1 optional-only; 0 listing-only institutions.

Rutgers University

Course	Term	Role	Edition as cited
Rutgers University · 22-544-645_26-198-641	Fall2026	Recommended	2010

Database Concepts Pearson · 1 institution

David Kroenke et al.

0 required/main; 0 optional-only; 1 listing-only institutions.

SUNY Farmingdale

Course	Term	Role	Edition as cited
SUNY Farmingdale · BCS260	Undated	Listed in departmental outline	8th

Database Design, Query Formulation, and Administration Sage · 1 institution

Michael Mannino

1 required/main; 0 optional-only; 0 listing-only institutions.

UT Dallas

Course	Term	Role	Edition as cited
UT Dallas · MIS6326	Fall2026	Required	8th

Formerly Prospect Press; grouped under its current Sage publisher.

Database Management and Design MyEducator · 1 institution

Gary Allen; Gary Hansen; Robert Jackson

1 required/main; 0 optional-only; 0 listing-only institutions.

San Jose State University

Course	Term	Role	Edition as cited
San Jose State University · INFO242	Spring2022	Required	1st, 2022

Database Modeling and Design: Logical Design Elsevier / Morgan Kaufmann · 1 institution

Toby Teorey; Sam Lightstone; Tom Nadeau

0 required/main; 1 optional-only; 0 listing-only institutions.

University at Buffalo

Course	Term	Role	Edition as cited
University at Buffalo · MGS404	Fall2009	Optional	4th

Database Principles: Fundamentals of Design, Implementation, and Management (ambiguous citation) Cengage · 1 institution

Coronel; Morris; Rob, as cited

0 required/main; 1 optional-only; 0 listing-only institutions.

SUNY New Paltz

Course	Term	Role	Edition as cited
SUNY New Paltz · CPS593	Summer2020	Optional / instructor source	10th, as cited

The syllabus title/edition combination is ambiguous. Kept separate from Database Systems rather than silently merging the two. A complete matching outline is not verified.

Database Processing: Fundamentals, Design, and Implementation Pearson · 1 institution

David Kroenke; David Auer et al.

0 required/main; 1 optional-only; 0 listing-only institutions.

SUNY New Paltz

Course	Term	Role	Edition as cited
SUNY New Paltz · CPS593	Summer2020	Optional / instructor source	12th

Database Systems: An Application-Oriented Approach, Introductory Version Pearson · 1 institution

Michael Kifer; Arthur Bernstein; Philip Lewis

1 required/main; 0 optional-only; 0 listing-only institutions.

Stony Brook University

Course	Term	Role	Edition as cited
Stony Brook University · CSE305	Fall2026	Listed main text	2nd
Stony Brook University · CSE305-532	Spring2021	Reference	2nd

GO! with Microsoft Access 2007, Volume 1 Pearson · 1 institution

Shelley Gaskin; Suzanne Marks; Kris Townsend

1 required/main; 0 optional-only; 0 listing-only institutions.

SUNY Canton

Course	Term	Role	Edition as cited
SUNY Canton · CITA104	Undated	Required	2007 / Volume 1

High Performance MySQL: Optimization, Backups, and Replication O’Reilly Media · 1 institution

Baron Schwartz; Peter Zaitsev; Vadim Tkachenko

0 required/main; 0 optional-only; 1 listing-only institutions.

SUNY Farmingdale

Course	Term	Role	Edition as cited
SUNY Farmingdale · BCS390	Undated	Listed in departmental outline	3rd

Information Systems for Business and Beyond Open educational resources · 1 institution

David Bourgeois et al.

0 required/main; 0 optional-only; 1 listing-only institutions.

University at Buffalo

Course	Term	Role	Edition as cited
University at Buffalo · MGS351	Fall2026-as-labeled	Listed course material	2019

Interactive SQL (OWL online text) Publisher not verified · 1 institution

Not identified in syllabus

1 required/main; 0 optional-only; 0 listing-only institutions.

UMass Amherst

Course	Term	Role	Edition as cited
UMass Amherst · CS197Q	Undated	Listed course text	Not stated

OWL is the delivery system. The syllabus does not establish the book’s commercial publisher or exact catalog identity.

Learn Python 3 the Hard Way Pearson · 1 institution

Zed A. Shaw

0 required/main; 1 optional-only; 0 listing-only institutions.

NYU Stern

Course	Term	Role	Edition as cited
NYU Stern · INFO-GB2346	Spring2018	Optional reference	Not stated

Addison-Wesley imprint; the saved syllabus describes online access as free, a historical claim not verified here.

Learning MySQL O’Reilly Media · 1 institution

Seyed Tahaghoghi; Hugh Williams

0 required/main; 1 optional-only; 0 listing-only institutions.

NYU Stern

Course	Term	Role	Edition as cited
NYU Stern · INFO-GB2346	Spring2018	Optional reference	Not stated; chapters 4-7

Learning Spark: Lightning-Fast Big Data Analytics O’Reilly Media · 1 institution

Jules Damji; Brooke Wenig; Tathagata Das; Denny Lee

0 required/main; 1 optional-only; 0 listing-only institutions.

Rutgers University

Course	Term	Role	Edition as cited
Rutgers University · 22-544-645_26-198-641	Fall2026	Recommended	2nd

Microsoft Access 365 Tutorial and Lab Manual Open educational resources · 1 institution

David Murray

0 required/main; 0 optional-only; 1 listing-only institutions.

University at Buffalo

Course	Term	Role	Edition as cited
University at Buffalo · MGS351	Fall2026-as-labeled	Listed course material	2024

Microsoft Office 365 Access 2016 Intermediate Cengage · 1 institution

Lisa Friedrichsen

0 required/main; 1 optional-only; 0 listing-only institutions.

SUNY Tompkins Cortland

Course	Term	Role	Edition as cited
SUNY Tompkins Cortland · CAPS131	Master2025-26	Recommended alternative	1st, 2016

Murach’s Oracle SQL and PL/SQL for Developers Mike Murach & Associates · 1 institution

Joel Murach

1 required/main; 0 optional-only; 0 listing-only institutions.

UT Austin

Course	Term	Role	Edition as cited
UT Austin · MIS325-BAX327	Spring2024	Required	2nd

MySQL Reference Manual O’Reilly Media · 1 institution

Michael Widenius; David Axmark; MySQL AB

0 required/main; 1 optional-only; 0 listing-only institutions.

Hofstra University

Course	Term	Role	Edition as cited
Hofstra University · CSC056	Fall2005	Reference	2002

New Perspectives Microsoft Office 365 & Access 2016 Comprehensive Cengage · 1 institution

Mark Shellman; Sasha Vodnik

0 required/main; 1 optional-only; 0 listing-only institutions.

Baruch College

Course	Term	Role	Edition as cited
Baruch College · CIS9340	Spring2025-Draft	Optional alternative	2016

New Perspectives Microsoft Office 365 & Access 2019 Comprehensive Cengage · 1 institution

Mark Shellman; Sasha Vodnik

0 required/main; 1 optional-only; 0 listing-only institutions.

Baruch College

Course	Term	Role	Edition as cited
Baruch College · CIS3400	Fall2026-Draft	Optional	2019 or 2021, as cited
Baruch College · CIS9340	Spring2025-Draft	Optional alternative	2019

New Perspectives Microsoft Office Access 2003, Introductory Cengage · 1 institution

Joseph Adamski; Kathleen Finnegan

1 required/main; 0 optional-only; 0 listing-only institutions.

Michigan Dearborn

Course	Term	Role	Edition as cited
Michigan Dearborn · MIS321	Winter2007	Required	2003 / introductory

Next Generation Databases: NoSQL, NewSQL, and Big Data Apress / Springer Nature · 1 institution

Guy Harrison

1 required/main; 0 optional-only; 0 listing-only institutions.

Rutgers University

Course	Term	Role	Edition as cited
Rutgers University · 22-544-645_26-198-641	Fall2026	Required	2016

NoSQL with MongoDB in 24 Hours, Sams Teach Yourself Pearson · 1 institution

Brad Dayley

0 required/main; 1 optional-only; 0 listing-only institutions.

UT Dallas

Course	Term	Role	Edition as cited
UT Dallas · ITSS4380	Spring2025	Suggested	2015

OCA Oracle Database 12c Installation and Administration Exam Guide (Exam 1Z0-062) McGraw Hill · 1 institution

John Watson

0 required/main; 0 optional-only; 1 listing-only institutions.

SUNY Farmingdale

Course	Term	Role	Edition as cited
SUNY Farmingdale · BCS390	Undated	Listed in departmental outline	Not stated

Oracle 9i: SQL with an Introduction to PL/SQL Cengage · 1 institution

L. L. Morris-Murphy

1 required/main; 0 optional-only; 0 listing-only institutions.

Michigan Dearborn

Course	Term	Role	Edition as cited
Michigan Dearborn · MIS322	Winter2007	Required	2003

Oracle Database 11g: The Complete Reference McGraw Hill · 1 institution

Kevin Loney

0 required/main; 1 optional-only; 0 listing-only institutions.

San Jose State University

Course	Term	Role	Edition as cited
San Jose State University · INFO242	Spring2022	Recommended	2009

Oracle Database 12c SQL McGraw Hill · 1 institution

Jason Price

0 required/main; 0 optional-only; 1 listing-only institutions.

SUNY Farmingdale

Course	Term	Role	Edition as cited
SUNY Farmingdale · BCS360	Undated	Listed in departmental outline	1st

Oracle SQL by Example Pearson · 1 institution

Alice Rischert

0 required/main; 1 optional-only; 0 listing-only institutions.

University at Buffalo

Course	Term	Role	Edition as cited
University at Buffalo · MGS613	Fall2012	Supplemental reference	4th

PHP and MySQL for Dynamic Web Sites: Visual QuickPro Guide Pearson · 1 institution

Larry Ullman

0 required/main; 1 optional-only; 0 listing-only institutions.

University at Buffalo

Course	Term	Role	Edition as cited
University at Buffalo · LIS569	Spring2012	Recommended	4th

Peachpit imprint.

Programming PHP O’Reilly Media · 1 institution

Rasmus Lerdorf; Kevin Tatroe

0 required/main; 1 optional-only; 0 listing-only institutions.

Hofstra University

Course	Term	Role	Edition as cited
Hofstra University · CSC056	Fall2005	Reference	2002

Python for Everybody: Exploring Data in Python 3 Open educational resources · 1 institution

Charles Severance

0 required/main; 1 optional-only; 0 listing-only institutions.

NYU Stern

Course	Term	Role	Edition as cited
NYU Stern · INFO-GB2346	Spring2018	Optional reference	Not stated

SQL Database Programming Questing Vole Press · 1 institution

Chris Fehily

0 required/main; 1 optional-only; 0 listing-only institutions.

NYU Stern

Course	Term	Role	Edition as cited
NYU Stern · TECH-UB26	Spring2026	Optional	Not stated

The syllabus does not identify an edition. The author’s current publisher is Questing Vole Press; the authorized online text is freely readable.

SQL for Data Scientists: A Beginner’s Guide for Building Datasets for Analysis Wiley / zyBooks · 1 institution

Renee M. P. Teate

0 required/main; 1 optional-only; 0 listing-only institutions.

UT Dallas

Course	Term	Role	Edition as cited
UT Dallas · ITSS4300	Spring2026	Suggested	Not stated

SQL in 10 Minutes, Sams Teach Yourself Pearson · 1 institution

Ben Forta

0 required/main; 0 optional-only; 1 listing-only institutions.

SUNY Farmingdale

Course	Term	Role	Edition as cited
SUNY Farmingdale · BCS260	Undated	Listed in departmental outline	4th

Seven Databases in Seven Weeks: A Guide to Modern Databases and the NoSQL Movement Pragmatic Bookshelf · 1 institution

Luc Perkins; Eric Redmond; Jim Wilson

1 required/main; 0 optional-only; 0 listing-only institutions.

Rutgers University

Course	Term	Role	Edition as cited
Rutgers University · 22-544-645_26-198-641	Fall2026	Required	2nd

Streaming Systems: The What, Where, When, and How of Large-Scale Data Processing O’Reilly Media · 1 institution

Tyler Akidau; Slava Chernyak; Reuven Lax

1 required/main; 0 optional-only; 0 listing-only institutions.

Rutgers University

Course	Term	Role	Edition as cited
Rutgers University · 22-544-645_26-198-641	Fall2026	Required	2018

The Data Warehouse Toolkit Wiley / zyBooks · 1 institution

Ralph Kimball; Margy Ross

0 required/main; 1 optional-only; 0 listing-only institutions.

Baruch College

Course	Term	Role	Edition as cited
Baruch College · CIS4400	Fall2026-Draft	Recommended	3rd
Baruch College · CIS9440	Undated-Draft	Recommended	3rd

Catalogue counts are retained from the prior audit and remain separate from the deduplicated platform analysis. The catalogue comes from the author's textbook adoption and contents audit [S4].

Student costs and course access

Price is part of the adoption conversation. These examples show the spread between a digital book, a printed text and a courseware package.

Existing product / format	Listed U.S. price	Access or format
Database Systems 15e eBook	US$71.99	Access through January 22, 2027

Existing product / format	Listed U.S. price	Access or format
Database Systems 15e eBook	US$98.99	Access through July 21, 2027
Database Systems 15e paperback	US$272.95	Print purchase
MindTap for Database Systems 15e	From US$149.00	Digital courseware; product / term selection applies

U.S. list prices retrieved October 25, 2026. Formats, terms and institutional agreements affect what students pay. No student price has been set for this proposal.

The proposed courseware

One connected learning journey: from understanding a business question to building the database, analyzing the evidence and making a decision.

An invitation to develop this together
A working proposal, open to better ideas.This outline is a draft and a suggestion, informed by the research and my classroom experience. I would like to work with the editors and reviewers to finalize the sequence, depth, tools and running projects. It is a starting point for our conversation.

Read, build, try, reflect

Each chapter introduces a business question, develops the core concepts, and includes a guided **Let's Build** , review and reflection questions, a **Reading Assessment Test** , and an independent **interactive lab** . Terms and takeaways support review.

A project that grows with the student

Two running cases connect the chapter activities. A separate semester project moves through a proposal, a working database and a final dashboard with a decision memo. Review the cases ↗ · See the assessment plan ↗

The complete chapter outline

Sixteen chapters in five parts, with a preface, a midterm integration studio, a closing chapter on careers, and seven appendices. Every chapter section is visible below, followed by the guided practice, independent work and research connection.

Preface · Preface: Why This Book, and How It Works

Why it was written and how to use it.

• P.1 Why I Wrote This Book

• P.2 How This Book Is Different

• P.3 How a Chapter Works

• P.4 The Tools, and Why They Were Chosen

• P.5 Two Routes Through the Book

• P.6 How AI Is Used in This Book

Part I · Chapters 1 to 3

From Data to Decisions

Main goal: Connect data to business performance, classify data correctly and organize information.

Chapter 1 · From Data to Business Performance

How does a business turn everyday activity into better results?

• 1.1 What a Business Is and How It Creates Value

• 1.2 Business Performance, Goals and KPIs

• 1.3 From Data to Wisdom: The DIKW Pyramid

• 1.4 Information Behavior: How People Seek and Use Information

• 1.5 Information Systems and Information Technology

• 1.6 The R.E.A.D. Framework: Recognize, Extract, Analyze, Decide

• 1.7 AI in the Data-to-Decision Chain

• 1.8 The Running Cases and the Semester Project

Let's Build: Trace one KPI back to the records that produce it.

Interactive lab: A KPI audit of a small business: three KPIs, the data each needs, and who decides with it.

Research connection: None of the four compared texts opens with business performance and KPIs. Business-systems framing appears in Buffalo MGS 351, Syracuse IST 195 and UNLV MIS 766.

Chapter 2 · Understanding Data

What kind of data is this, and what can I legitimately do with it?

- 2.1 Data and Information
- 2.2 Digital and Analog Data; Bits, Bytes and Tokens
- 2.3 Data Types: Numbers, Text, Dates, Booleans and Files
- 2.4 Qualitative and Quantitative Data
- 2.5 Levels of Measurement: Nominal, Ordinal, Interval, Ratio
- 2.6 Structured, Semi-Structured and Unstructured Data: CSV and JSON
- 2.7 What Makes a Good Table
- 2.8 Data Quality
- 2.9 From Data to Insight: What Each Type Allows

Let's Build: Profile a dataset column by column: type, scale and allowed operations.

Interactive lab: Data Type Detective: classify and repair a messy business file, then draw one valid insight per column type.

> **Research connection:** The compared outlines do not give data literacy and measurement scales their own opening chapter. Data preparation appears in NYU TECH-UB 26 and Temple MIS 2502.

Chapter 3 · Organizing Data and Information

How do we make data findable, reusable and trustworthy?

- 3.1 Why Organization Matters
- 3.2 Files, Folders, Naming and Versions
- 3.3 Hierarchies, Tags and Facets
- 3.4 Metadata, with YAML as an Example
- 3.5 Taxonomies and Controlled Vocabularies
- 3.6 Data Dictionaries
- 3.7 Spreadsheets as Tables, and Where They Break
- 3.8 From Organized Files to Databases

Let's Build: A data dictionary and a folder-and-tag scheme for the running case.

Interactive lab: Reorganize and document a shared-drive mess.

Research connection: Added by design from information science, so students see why databases exist before they build one. Not a standalone topic in the syllabi reviewed.

Part II · Chapters 4 to 6

Databases and SQL

Main goal: Read a relational database and answer business questions with SQL.

Chapter 4 · Databases and the Relational Model

Why do organizations keep data in related tables instead of files?

- 4.1 What a Database Is and What a DBMS Does
- 4.2 Local and Cloud Databases: SQLite, PostgreSQL, Supabase, MySQL, Access
- 4.3 Tables, Rows, Columns and Data Types
- 4.4 Primary, Foreign and Composite Keys
- 4.5 Relationships: One-to-One, One-to-Many, Many-to-Many
- 4.6 Schemas, Integrity Rules and Constraints
- 4.7 Database Objects: Tables, Queries, Views, Forms and Reports

Let's Build: Explore a supplied case database in Supabase Studio and trace a relationship.

Interactive lab: Read an unfamiliar schema and predict the answers to five business questions before querying.

Research connection: Design and data modeling are named in all 13 catalogue benchmark descriptions; relational foundations are documented in Baruch CIS 3400, George Mason MIS 310 and SUNY Canton CITA 215.

Chapter 5 · SQL Foundations

How do I ask one table a precise question?

• 5.1 SELECT and FROM

• 5.2 Filtering with WHERE, LIKE, IN and BETWEEN

• 5.3 Sorting, DISTINCT and LIMIT

• 5.4 Working with NULL and Dates

• 5.5 Aggregate Functions

• 5.6 Creating Tables and Inserting Data

• 5.7 Safe UPDATE and DELETE

• 5.8 AI verification: Accept, Revise or Reject an AI-Written Query

Let's Build: Answer ten business questions on the running case.

Interactive lab: The same skills on the second case, with an AI verification.

Research connection: SQL appears in 10 of 13 catalogue benchmark descriptions (77%) and in 52.9% of the 855 analyst postings. Early SQL in Rutgers 33:136:470, Northeastern CS 3200 and UT Dallas ITSS 4300.

Chapter 6 · SQL for Business Analysis

How do I combine tables into trustworthy business measures?

• 6.1 INNER and LEFT Joins

• 6.2 Joining Three or More Tables

• 6.3 GROUP BY and HAVING

• 6.4 CASE Expressions for Business Categories

• 6.5 Calculated Measures: Rates, Averages and Shares

• 6.6 Grain and the Double-Count Problem

• 6.7 Checking Results by Hand

Let's Build: Build a small KPI report from joined tables.

Interactive lab: Find and fix a double-counted measure.

Research connection: Joins and aggregation are where business reports most often go wrong. Taught at Rutgers, Temple, UT Austin and UT Dallas.

Midterm studio · Integration Studio A (midterm)

An unnumbered review week between Parts II and III.

Cumulative review of Chapters 1 to 6, a practice exam, and **Project Checkpoint 1** : proposal and data plan.

Research connection: Staged, multi-phase projects recur across the course documents reviewed, so the project has graded checkpoints through the semester.

Part III · Chapters 7 to 11

Designing and Running Databases

Main goal: Design, normalize, build, test and secure a database of their own.

Chapter 7 · Database Design: From Requirements to ER Diagrams

How do business rules become a database design?

• 7.1 The Systems Development Life Cycle

• 7.2 Waterfall and Agile

• 7.3 Requirements and Business Rules

• 7.4 Entities, Attributes and Relationships

• 7.5 Cardinality, Optionality and Crow's Foot Notation

• 7.6 Associative Entities and Many-to-Many Relationships

• 7.7 Drawing ERDs with Lucidchart, diagrams.net or Mermaid

Let's Build: Draw the ERD for the running case from its business rules.

Interactive lab: Design an ERD for a new business process.

Research connection: Design and data modeling are named in all 13 catalogue benchmark descriptions. ER modeling comes before normalization, as in Coronel and Morris, Kroenke and Hoffer.

Chapter 8 · Normalization

How do we design out errors before they happen?

• 8.1 Update, Insert and Delete Anomalies

• 8.2 Functional Dependencies

• 8.3 First Normal Form

• 8.4 Second Normal Form

• 8.5 Third Normal Form

• 8.6 Decomposing and Rejoining Tables

• 8.7 When to Denormalize

Let's Build: Normalize a flat spreadsheet to third normal form.

Interactive lab: Normalize a real business form and prove no data was lost.

Research connection: Normalization is core in each of the four compared texts and is documented in Baruch, Rutgers, George Mason, Georgia Tech and SUNY Farmingdale BCS 260.

Chapter 9 · Building and Testing a Database

How do we turn a design into a working, trustworthy database?

• 9.1 Data Definition Language: CREATE, ALTER, DROP

• 9.2 Constraints

• 9.3 Loading Data from CSV

• 9.4 Staging and Cleaning

• 9.5 Integrity Tests

• 9.6 Data Entry Workflows and Forms

Let's Build: Build and load the running-case database in Supabase.

Interactive lab: Build the lab case from its ERD and run an integrity test suite.

Research connection: Hands-on building appears in 10 of 13 catalogue benchmark descriptions (77%). Implementation projects at Rutgers, Baruch and Boston University MET CS 669.

Chapter 10 · Advanced SQL

How do analysts write queries that are reusable and easy to check?

- 10.1 Subqueries
- 10.2 Common Table Expressions
- 10.3 Views
- 10.4 Conditional Aggregation
- 10.5 Window Functions
- 10.6 Set Operations
- 10.7 Functions and Triggers
- 10.8 Reading a Query Plan with EXPLAIN

Let's Build: A reusable analysis layer of views and CTEs.

Interactive lab: A monthly performance report with rankings and period-over-period change.

Research connection: Analytical SQL with CTEs and window functions appears in current syllabi at NYU Stern and UT Dallas.

Chapter 11 · Database Administration, Security and Reliability

How do we keep business data safe, correct and available?

- 11.1 Roles, Users and Least Privilege
- 11.2 Row-Level Security
- 11.3 Transactions and ACID
- 11.4 Concurrency
- 11.5 Backup and Recovery
- 11.6 Indexes and Performance

• 11.7 Governance, Privacy and Compliance

• 11.8 Audit Trails

Let's Build: Add roles, row-level security and an audit trigger to the running case.

Interactive lab: A security and recovery plan for the lab case. Project Checkpoint 2: working database.

Research connection: Administration, security or transactions appear in 6 of 13 catalogue benchmark descriptions (46%), including Rutgers, UT Austin, George Mason and SUNY Canton.

Part IV · Chapters 12 to 15

From Data to Insight

Main goal: Analyze results, visualize them, report in Power BI and work with warehouses and AI.

Chapter 12 · Business Analytics and Decision-Making

What does the data actually tell us, and how sure are we?

• 12.1 Four Kinds of Analytics

• 12.2 Framing an Analytical Question

• 12.3 Distributions, Segments and Fair Comparisons

• 12.4 Correlation and Causation

• 12.5 Uncertainty

• 12.6 From Measure to Decision

• 12.7 The Decision Memo

Let's Build: A SQL analysis that answers one management question.

Interactive lab: An analytical memo for the lab case.

Research connection: Closes the loop from query to decision with a decision memo. Business analysis tasks appear in Temple MIS 2502 and Georgia MIST 4610.

Chapter 13 · Reporting and Data Visualization

How do we show results so people decide well?

- 13.1 Reports and Dashboards
- 13.2 Choosing the Right Chart
- 13.3 Honest and Misleading Visuals
- 13.4 Accessible Design
- 13.5 Designing for Decision-Makers
- 13.6 The Tableau Route

Let's Build: Redesign a misleading dashboard.

Interactive lab: A one-page dashboard specification.

> **Research connection:** Only 4 of 13 catalogue benchmark descriptions (31%) include BI or analytics. Visualization appears in Georgia MIST 4610 and UT Austin MIS 325.

Chapter 14 · Power BI

How do we turn a database into an interactive business report?

- 14.1 Connecting to a Database
- 14.2 Shaping Data with Power Query
- 14.3 The Data Model and Relationships
- 14.4 Measures, Calculated Columns and DAX Basics
- 14.5 Building an Interactive Report
- 14.6 Validating Every Number Against SQL

Let's Build: A Power BI report on the running case.

Interactive lab: A Power BI report on the lab case, validated against SQL.

> **Research connection:** Power BI is listed in 29% of the 855 analyst postings and is not documented in any of the course families reviewed, so it is added by design.

Chapter 15 · Big Data, Data Warehouses and AI

How do organizations work with data at scale, and where does AI fit?

- 15.1 OLTP and OLAP
- 15.2 Warehouses, Marts, Lakes and Lakehouses
- 15.3 Star Schemas: Facts and Dimensions
- 15.4 ETL, ELT, APIs and JSON
- 15.5 Big Data and NoSQL
- 15.6 AI and Data: Text-to-SQL, Embeddings and Retrieval
- 15.7 Verifying AI Output

Let's Build: A small star schema from the running case.

Interactive lab: Evaluate three AI answers about the lab data: accept, revise or reject.

Research connection: The 13-course catalogue benchmark names AI in one description and cloud computing in none. Warehousing appears in Baruch CIS 4400 and graduate AI topics in UT Dallas BUAN 6V99; these inform extensions, not universal requirements.

Part V · Chapter 16 and Closing

Putting It to Work

Main goal: Deliver a capstone decision package and plan a path into data careers.

Chapter 16 · Capstone: From Data to Decisions

Can you take a real business problem all the way to a defensible decision?

- 16.1 Scoping the Problem and the KPI
- 16.2 From Data Dictionary to ERD
- 16.3 Building the Database
- 16.4 Analysis and a Validated Dashboard
- 16.5 The Decision Memo
- 16.6 Demonstration and Reflection

Project Checkpoint 3: the final project, with a working database, analysis, dashboard and decision memo, and an individual demonstration.

> **Research connection:** Staged projects at Rutgers, Boston University and Georgia. The individual demonstration responds to the uneven AI policies documented in the syllabi.

Closing chapter · Closing: Where Data Can Take You

Where do these skills lead?

- C.1 The Data-to-Decision Journey on One Page
- C.2 Careers in Data: Analyst, BI Analyst, Database Administrator, Data Architect, Data Engineer
- C.3 Data Skills in Every Business Major
- C.4 Building a Portfolio

> **Research connection:** The labor and employer research above: data scientist jobs growing 35% and database architect jobs 9% through 2035.

Appendices

These companions keep setup steps and platform alternatives close at hand without interrupting the main story. Instructors can choose the tools their course needs, while students follow the same core concepts and cases.

A

Supabase and the In-Book SQL Editor

Setting up a Supabase project, using Supabase Studio, and (planned) running PostgreSQL inside the reader.

B

MySQL Route

Planned: schemas, seed data and solutions for every SQL chapter, to be tested on the shared cases, for courses built on MySQL.

C

Microsoft Access Route

Tables, relationships, visual queries, forms and reports on the same cases. An Access appendix and Access labs exist today.

D

SQL Reference and Dialect Guide

PostgreSQL, MySQL, SQL Server and SQLite side by side.

E

Diagramming with Lucidchart

ER diagrams in Lucidchart, with diagrams.net and Mermaid as free alternatives.

F

Tableau Route

The visualization chapter's activities in Tableau.

G

Aa

Terms Treasury

The glossary of every key term in the book.

How students progress

• **Understand the data** Chapters 1 to 3: what data is, how it measures performance, and how to organize it.

• **Use a database** Chapters 4 to 6: query a supplied database before designing one.

• **Build one** Chapters 7 to 11: design, normalize, build and secure their own database.

• **Decide and defend** Chapters 12 to 16: turn results into dashboards and a decision memo they can explain and defend.

Two ways to use the book

Both routes reflect the syllabus research: business-core courses emphasize data use, design and reporting, while MIS courses need the full database sequence. Both are selectable; final pacing is to be agreed in review.

Business-core data course	Chapters 1 to 8, then 12 to 14.
MIS database course	Chapters 1 and 4 to 16, with Chapters 2 and 3 as assigned reading.

Technology and tools

The syllabus platform audit distinguishes documented practical use, a named option, and a software-list mention. Counts use 54 selected course families and overlap across tools. Official documentation then informs the proposed PGlite, Supabase, and Power BI pathway. [S2] [T1] [T2] [T3]

Practical use means the reviewed text assigns or demonstrates work with the tool; an option does not establish that a student selected it. Twenty course families lack sufficiently explicit practical / listed platform evidence. No adoption of Supabase or PGlite was established in this archive.

Main conclusion

MySQL and Access have substantial documented footprints in the reviewed courses. Keep meaningful compatibility routes while using one principal SQL foundation. The choice of PostgreSQL is a design recommendation, rather than the dominant platform in this sample.

The proposed technology pathway

Recommendation: make **PostgreSQL** the SQL foundation, with **PGlite** for immediate browser practice and **Supabase** for the hosted project. Use **Power BI** as the main reporting studio, **Tableau** as an alternative, and **Lucidchart** for modeling with a supported alternative. Keep the existing **Microsoft Access** route; develop and test a matching **MySQL** route.

Layer	Recommended role	Condition for adoption
SQL	Main skill and reasoning language.	Portable core examples with PostgreSQL-specific syntax labeled.
PGlite	Immediate embedded practice.	Planned. Implement and test persistence, reset, accessibility and known-answer checks.
Supabase	Hosted project and operational controls.	Classroom onboarding, export/restore, permission tests and a supported exit route; no guaranteed free service claim.
Power BI	Principal BI studio in Chapter 14.	Windows authoring access or a supported institutional virtual lab.

Layer	Recommended role	Condition for adoption
Tableau	Equivalent alternate visualization studio.	Same data, measures and rubric; avoid two required BI tools in the standard route.
Lucidchart	Diagramming example.	Provide a supported alternative; no presumed free-tier entitlement.
Access / MySQL	Compatibility routes. Access is a route in use (labs and an appendix); MySQL is a planned route.	Test the MySQL route on the shared cases, with equivalent core examples and instructor setup; not every student must learn both.

PGlite executes locally in the browser; Supabase is hosted PostgreSQL. Export / import must be taught explicitly. Power BI Desktop requires Windows or an institutional virtual lab. “Browser-first SQL practice with a hosted project” accurately describes the intended path; integration, access, licensing, and maintenance still need implementation decisions.

Companions and appendices

Setup, alternative routes and reference material support the tools each instructor chooses.

Companion	Purpose
PostgreSQL / PGlite / Supabase	Setup, workspace reset/persistence, portable SQL, explicit transfer, exports, restore and access tests.
MySQL / MySQL Workbench	A planned alternate route, to be tested with matching data, query results and dialect differences. An appendix-length syntax list alone is insufficient.
Microsoft Access	In use today. Tables, relationships, visual queries, forms, reports and an equivalent small application.
Lucidchart and alternatives	Crow’s Foot modeling, exports and review; diagrams.net/ERDPlus or Mermaid alternatives avoid a mandatory subscription.
Power BI and Tableau	A complete Power BI studio plus an equivalent Tableau reporting route; Mac access planning is explicit.
SQL reference and advanced database topics	Portable core syntax, dialect flags, triggers/procedures, deeper windows and normal forms.
Capstone, reviews and instructor resources	Milestones, tests, rubrics, solution files, slides, question banks and course-route maps.

Running projects

Running cases

Every Let's Build uses one guided case and every lab uses a second case, so students learn a skill once and then transfer it. The Grading Database and PetVax are the existing running cases. A small retail business is proposed as an alternative for evaluation; it is not a completed third case.

Guided case · in use

Grading Database

Courses, assignments and grades offer a familiar setting, although policy complexity still needs careful scoping.

What to test: Keep as the guided case with a small, stable schema.

Lab case · in use

PetVax veterinary clinic

Customers, pets, visits, services and payments give real relationships and operating KPIs.

What to test: Whether students relate to the setting.

Proposed alternative · for evaluation

Small retail business

Orders, products, customers and revenue suit the analytics, Power BI and star-schema chapters.

What to test: Whether the extra entities overload an introductory lab.

Learning and assessment

Each chapter uses the components listed in the overview. Here is what each one assesses.

Assessment that makes reasoning visible

Students predict and define before execution, test keys and totals during execution, and interpret the result afterward. Reading checks, guided practice, independent labs, cumulative reviews, and the final project serve distinct purposes.

Assessment	Purpose	Proposed evidence
Reading assessment tests	Check readiness and conceptual distinctions.	Short scenario, output-prediction and misconception items; separate formative feedback from graded evidence.
Let’s Build	Guided skill development.	Worked reasoning, expected results and checkpoints.
Interactive chapter labs	Independent transfer.	SQL, numeric answers, diagrams, files and short interpretations, appropriate to the chapter.
Checking AI-assisted work	Practice judging AI-assisted work, from Chapter 5 on.	Accept, revise or reject an AI-written query or answer, with a short written reason.
Project checkpoints	Stage the semester project.	Proposal and data plan at the Midterm Integration Studio, a working database in Chapter 11, and the final project in Chapter 16.
Cumulative tests / reviews	Integrate ideas across chapters.	Unfamiliar schema or changed business rule; explain and repair.
Final project	Demonstrate a complete usable solution.	Rebuildable database, validation log, report, decision memo and individual demonstration.

Assessment principle

Use known-result SQL and integrity tests where they fit; use human review for modeling, explanations, and business judgment. Evaluate unfamiliar variations and require a short individual explanation. Automated feedback on SQL and lab answers is a prototype in testing, and AI-assisted feedback is a development option that needs validation.

One problem. Three checkpoints. A defensible recommendation.

• **Question and requirements.** A scoped business problem, stakeholder, measure, source plan and acceptance criteria. Start in the opening chapters.

• **Design and working data.** An ERD, normalization rationale, implemented schema, synthetic data, import/reject log and integrity checks.

• **Evidence and action.** Tested SQL, reconciled reporting, essential access/recovery controls, a decision memo and an individual change task.

Proposed rubric	Weight
Problem framing and requirements	10%
Data model and normalization rationale	20%
Implementation and data quality	20%
SQL correctness and verification	20%
Reporting and business interpretation	15%
Reproducibility, controls and individual explanation	15%

Rubric weights are a proposal for faculty review, not a change to the live course. A compact submission includes schema.sql, seed data, queries.sql, an ERD, validation evidence, README and a short decision memo. A live link alone is insufficient because its data may change after submission.

How it is different

This book differentiates itself by teaching business students to use data to make decisions. It differs from the main textbooks by:

• **Starting with the business.** Chapter 1 opens with performance, goals and KPIs, before any database theory.

• **Teaching data before databases.** Data types, measurement scales and organizing information come first.

• **Using a database before designing one.** Students query a supplied database in Chapters 4 to 6, then design their own.

• **Ending with the decision.** Analytics, dashboards and a full Power BI chapter lead to a decision memo.

• **One SQL dialect, flexible tools.** SQL is taught in the PostgreSQL dialect, with a portable core and explicit dialect differences. The Access route is in use today; a MySQL route is planned and will be tested on the same cases.

• **Industry tools students can keep.** Supabase hosts the PostgreSQL project database; Power BI is the main reporting studio, with Tableau as an alternate route. Classroom accounts, licensing and access need an agreed support plan.

• **Making AI a habit.** From Chapter 5 on, an AI verification asks students to accept, revise or reject AI-assisted work and to take responsibility for the result.

• **Practice in every chapter.** A Let's Build, an interactive lab and an auto-graded quiz, all on the book's own cases.

• **Developed in class.** Version 1 was tested in summer 2026; version 2 is in use with 500 students in fall 2026.

Advantages of this courseware

The starting point is a complete course already in use. The opportunity is to develop it with editorial support, a clear tool pathway, and wider instructor review.

Ready asset	The interactive textbook, chapter practice, quizzes and instructor resources are in use with 500 students this semester. The next edition builds on that foundation.
Two course routes	A full applied MIS database route and a shorter business-core route, selected from the same chapters.
Portfolio fit	A business-focused pathway from data meaning to SQL, reporting and a decision memo. Positioning alongside existing database titles is a conversation to have together.
Known partner	I already work with Cengage as a member of its Intro Computing Thought Leadership Board.
Additional opportunities	Selected modules may also fit accounting information systems and analytics foundations. Existing Access materials and a planned MySQL route can support different teaching environments.

Open questions

Three areas to decide together.

• 1Publishing arrangement and portfolio positioningThe form of the publishing relationship, and where the book sits among Cengage's current titles.

• 2Development scopeThe tools and routes to support, the running cases, platform fit, and the sample unit reviewers see first.

• 3Responsibilities, review milestones and manuscript submissionWho does what, what reviewers receive and when, and the manuscript submission timeline.

Detailed agenda The questions raised in the earlier brief and prospectus, grouped under the three areas

1. Publishing arrangement and portfolio positioning

• **Positioning:** a new applied title, a successor to Concepts of Database Management, or a companion to Database Systems (Coronel and Morris)?

• **Primary market:** the applied MIS database course first, the business-core course, or both routes from one book?

• **Complement, not compete:** where the outline should complement Cengage's current titles, and where it should not compete.

2. Development scope, tools, cases and sample unit

• **Tools and access:** which routes to support fully (Access, MySQL, PostgreSQL), and how students reach Power BI.

• **Platform fit:** how the chapter package (reading assessments, labs, SQL practice) maps to MindTap, and whether a live SQL environment such as Supabase or in-browser PostgreSQL can connect to it.

• **Running cases:** keep the Grading Database and PetVax, or evaluate a commercial case such as retail orders and revenue?

• **Sample unit:** which complete unit (reading, guided build, lab, solution and rubric) reviewers should see first.

3. Responsibilities, review milestones and manuscript submission

• **Review package:** what reviewers receive (this outline, sample chapters, a lab, reader access), how many reviewers, and with what criteria.

• **Responsibilities:** who maintains the platform, the cases and the instructor materials during development.

• **Timeline:** the key review dates and the manuscript submission plan.

Suggested next steps

I would like to formalize our publishing relationship and agree on the development scope and manuscript submission timeline.

• **Let’s cement the publishing relationship.** Agree on the arrangement and the book’s place in the portfolio.

• **Agree on the development scope.** Settle the audience, tools, cases and the sample unit.

• **Set the milestones.** Agree on responsibilities, review milestones and the manuscript submission timeline.

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✦A final thought
This book is a labor of love, and of many, many hours of work. I would be honored to see it come to life as a published book, and to see other universities and instructors make it their own.
Nim

From a classroom question to a complete learning journey.

Using Data to Drive Business Performance

Explore the existing courseware and the materials behind the proposed next edition.

References

Market, employment and curriculum references

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• AACSB International. (2026, September 9). Two institutions earn AACSB initial business accreditation, September 2026 (1,098 accredited institutions in 71 countries and territories). Link

• 365 Data Science. (2026, April 28). The data analyst job market in 2026: Research on 1,000 job postings (Indeed USA; 855 relevant postings). Vendor analysis, not peer reviewed. Link

• U.S. Bureau of Labor Statistics. (2026). Occupational Outlook Handbook: Data scientists. Link

• U.S. Bureau of Labor Statistics. (2026). Occupational Outlook Handbook: Database administrators and architects. Link

• World Economic Forum. (2025). The Future of Jobs Report 2025. Link

• Cengage. Concepts of Database Management, 10th edition (2021). Link

Research archive, textbook, labor and platform references

[S1]

Author's syllabus archive and manifest

63 public course documents (syllabi, catalog descriptions and schedules) from 31 institutions, collected from institutional websites on October 1, 2026. Mixed course levels and dates; the original source of each document is linked in the course register. Archive available on request.

[S2]

Author's database-platform evidence model

October 17, 2026 audit of the same archive. Role-coded platform evidence (practical use, option, listed) for 54 selected course families; counts recalculated October 25, 2026. Available on request.

[S4]

Author's textbook adoption and contents audit

October 10, 2026 report: 60 named books and course texts in the archive, with school-level mentions. Historical mentions, not market shares. Available on request.

[P2]

Cengage: Database Systems, 15th edition ↗

Coronel and Morris. Official current listing checked October 25, 2026: 16 chapters in six parts, with Introduction to SQL (Chapter 4) and Advanced SQL (Chapter 5) before ER modeling; auto-graded database labs in MindTap.

[P3]

Cengage: Concepts of Database Management, 10th edition ↗

Friedrichsen, Ruffolo, Monk, Starks, Pratt and Last. Official publisher search results provide the nine-module outline and Access practice description.

[P4]

Pearson: Database Concepts, 10th edition ↗

Kroenke, Auer, Vandenberg and Yoder. Current publisher educator page; the uploaded PDF is the older ninth edition, not the tenth.

[P5]

Pearson: Modern Database Management, 14th edition ↗

Hoffer, Topi, Venkataraman and Bala. Official contents and description checked October 25, 2026: 13 chapters, SQL from Chapter 5, an analytics chapter and new State of AI sections.

[L1]

U.S. Bureau of Labor Statistics: Data Scientists ↗

Current Occupational Outlook Handbook read October 25, 2026. Projects 35% employment growth, 2025-2035; all occupations 3%.

[L2]

U.S. Bureau of Labor Statistics: Database Administrators and Architects ↗

Current Occupational Outlook Handbook read October 25, 2026. Architects 9%; administrators 0%; combined 4%, 2025-2035. Rounded percentages.

[L3]

World Economic Forum: Future of Jobs Report 2025 ↗

Employer expectations for 2025-2030: analytical thinking essential to about seven in ten responding companies; AI and big data lead skills expected to grow. Global employer survey, not U.S. hiring requirements.

[T1]

PGlite: What is PGlite? ↗

Official documentation: embedded PostgreSQL compiled to WebAssembly; runs in the browser, with optional local persistence. It is not itself a hosted cloud database.

[T2]

Supabase: Database overview ↗

Official documentation: each project provides a PostgreSQL database. Hosted operations, accounts, permissions and product limits require a separate teaching plan.

[T3]

Microsoft: Get Power BI Desktop ↗

Current installation documentation, checked October 25, 2026. Windows desktop authoring creates an access requirement for Mac users.